

# Decoupling to Fuse: Learning Shared and Modality-Specific Representations for Multi-Sensor 3D Object Detection

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## 1 Introduction

Accurate 3D object detection is essential for autonomous driving. Cameras, LiDAR, and 4D radar observe a scene through different sensing mechanisms, providing visual appearance, geometric structure, and radar-return information. Combining these observations can compensate for limitations of individual sensors and provide a richer basis for detecting objects. Realizing this potential requires integrating features with different sensing characteristics and spatial sampling patterns into a joint representation that supports detection, motivating continued research on multi-sensor fusion (Liu et al., 2023; Park et al., 2025b; Yue et al., 2025).

Existing approaches have advanced multi-sensor detection through spatial interaction and adaptive fusion. Establishing correspondence in a common spatial domain allows features from different sensors to exchange information about the same regions. Attention and gating mechanisms further adjust their contributions, while foreground modeling and reliability estimation help account for differences in input quality. Together, these developments provide an effective foundation for combining sensor observations under varying scene and sensing conditions (Chae et al., 2024; Huang et al., 2025; Chae et al., 2025; Paek and Kong, 2025; Park et al., 2026).

Nevertheless, spatial correspondence and adaptive weighting do not by themselves distinguish the information that can be shared across sensors from the cues specific to each modality. Even within aligned local regions, these two types of information remain mixed in the input features, although they play different roles in fusion: shared information can provide cross-modal support, while modality-specific cues can preserve complementary observations. Learning consistency alone may suppress useful differences, whereas emphasizing differences alone may overlook evidence shared across sensors. This motivates approaching fusion through representation decoupling: explicitly learning shared and modality-specific representations and using both to construct the fused detection features. Such a formulation builds on advances in learning and retaining common and distinct information for multimodal prediction (Liang et al., 2023; Liu et al., 2025; Qian et al., 2026; Wang et al., 2026).

For dense 3D detection, this decoupling also needs a clear connection to detection targets. Shared responses may originate from either objects or background, and modality-specific differences may reflect useful observations or noise. Separating the two types of information therefore does not by itself ensure their relevance to detection. We address this issue by grounding representation constraints in object regions and using object-related information to guide the contribution of the learned features to fusion. Object guidance thus gives the learning and use of decoupled representations a detection-specific focus.

We introduce **ObjDec**, a multi-sensor 3D detection architecture centered on shared and modality-specific representation decoupling. Within spatially corresponding bird’s-eye-view (BEV) patches,

separate mappings construct the two representations from each sensor token, and both participate in building the fused detection features. Object-region supervision encourages cross-modal agreement in shared representations, preserves modality-specific differences, and limits similarity between the two branches. The learned representations further determine feature updates, modality contributions, and object context, allowing their information to shape fusion inputs and cross-modal interaction. Foreground gating and context modulation implement this connection between decoupled representations and fusion computation. The representation mappings operate at every patch; object regions specify where the representation constraints are applied during training. Ground-truth boxes provide training supervision only, while inference predicts all fusion-control signals from input features without prior detections or weather labels.

We evaluate ObjDec primarily on K-Radar. At a confidence-filtering threshold of 0.3 on v1, ObjDec achieves 88.36%  $AP_{3D}$  at IoU 0.3 and 88.10%  $AP_{BEV}$  at IoU 0.5. Compared with the archived results associated with the released fusion baseline under the same filtering setting, these scores improve by 8.04 and 7.77 percentage points, respectively, with approximately 1.7% more parameters. Component ablations assess the contribution of each design, while representation visualizations and spatial gate analyses examine the learned feature organization and fusion behavior. We also evaluate a decoupled-control variant on K-Radar v2 and train and test the architecture on V2X-Radar-V, examining its applicability beyond the primary category and sensor setting under the specified dataset protocols (Paek et al., 2022; Paek and Kong, 2025; Yang et al., 2025).

Our main contributions are as follows:

- **A fusion architecture based on representation decoupling.** We develop a multi-sensor 3D detection architecture that explicitly learns shared and modality-specific representations in corresponding local regions and uses both to construct fused detection features.
- **Object-guided learning and use of decoupled representations.** We supervise representation decoupling within object regions and connect the learned representations to foreground gating, modality contribution control, and object context, guiding their use in detection-oriented fusion.
- **Evaluation of detection performance and model behavior.** We evaluate detection performance and component contributions on K-Radar, analyze representations and spatial gating, and examine the architecture under extended categories and target-dataset training on V2X-Radar-V.

**[TO COMPLETE: Insert the final author-approved motivation figure. This box only reserves its position.]**

**Figure 1:** Motivation for object-guided shared and modality-specific representation learning and fusion.

## 2 Related Work

### 2.1 Multi-Sensor Fusion for 3D Object Detection

Multi-sensor detectors combine observations through spatial correspondence and feature interaction. BEVFusion (Liu et al., 2023) brings camera and LiDAR features into a unified bird’s-eye-view (BEV) space, while TransFusion (Bai et al., 2022) and CMT (Yan et al., 2023) establish cross-modal interactions through LiDAR-guided queries and 3D positional encoding, respectively. For radar-camera fusion, RCBEVDet (Lin et al., 2024) combines radar BEV encoding with deformable cross-attention for feature alignment. These designs offer different ways to establish correspondence and integrate sensor features.

Recent studies further address modality contributions and varying input quality. MoME (Park et al., 2025b) routes queries among camera, LiDAR, and joint expert decoders. CCF (Wu et al., 2026) combines separate query supervision, geometric priors, and complementary masking to address modality imbalance in cross-domain detection. RobuRCDet (Yue et al., 2025) uses Gaussian expansion of radar points and camera-confidence-guided fusion to accommodate different sensor disturbances.

On K-Radar, 3D-LRF (Chae et al., 2024) and L4DR (Huang et al., 2025) investigate spatial interaction, foreground denoising, and gated fusion. DLRFFusion (Chae et al., 2025) separately encodes Doppler, radar power, and LiDAR features for iterative interaction. WCBR (Li et al., 2026) adjusts branch contributions according to environmental conditions, while SRF (Mukhatbekov et al., 2026) fuses stereo images and radar in 3D voxel space. For three-sensor fusion, ASF (Paek and Kong, 2025) uses unified canonical projection and cross-sensor attention within corresponding patches, whereas RAF (Park et al., 2026) modulates camera features using explicitly supervised image reliability maps. Building on ASF’s projection and local interaction mechanisms, ObjDec makes shared and modality-specific representation decoupling an explicit part of the local fusion architecture.

### 2.2 Shared and Modality-Specific Representation Learning

Explicitly organizing shared and specific information has an established foundation. Domain Separation Networks (Bousmalis et al., 2016) distinguish shared and domain-specific representations for domain adaptation. MISA (Hazarika et al., 2020) learns modality-invariant and modality-specific representations through similarity, difference, and reconstruction constraints for affect analysis. FactorCL (Liang et al., 2023) retains task-relevant shared and unique information, while DeCUR (Wang et al., 2024) learns common and unique representations through cross-modal and intra-modal correlation objectives. These studies support learning consistency while preserving useful differences.

Recent work further explores how these representations are selected, aligned, and used. Liu et al. (2025) use a causal formulation to further separate modality-specific information into unique and redundant components. MultiLoReFT (Tonekaboni et al., 2026) learns shared and modality-specific subspaces through low-rank representation fine-tuning. DecAlign (Qian et al., 2026) uses prototype-guided optimal transport and distribution matching for modality-specific and common representations, respectively. For detection, SPFD (Wang et al., 2026) separates RGB-event shared and private features through frequency-domain coherence and uses them in both its encoder and decoder. LMD (Park et al., 2025a) instead provides post-hoc decomposition of modality contributions in pretrained fusion networks.

Building on this line of work, ObjDec uses shared and modality-specific representation decoupling to organize local fusion for multi-sensor 3D detection. Its design connects spatially corresponding

sensor patches, representation constraints supervised within object regions, and direct use of the decoupled representations in token updates and cross-modal interaction. Object guidance gives the learning and use of these representations a detection-specific focus, while representation decoupling provides the organizing principle of the fusion architecture.

## 3 Method

### 3.1 Overview

We introduce ObjDec, a multi-sensor 3D detection architecture centered on shared and modality-specific representation decoupling. As illustrated in Figure 2, modality-specific encoders first transform camera, LiDAR, and 4D radar observations into spatially corresponding bird’s-eye-view (BEV) features. We adopt ASF’s unified projection to map features with different channel dimensions into patch tokens for corresponding local regions, and retain its local cross-sensor attention and subsequent feature transformation as the interaction backbone (Paek and Kong, 2025). Let  $\mathcal{M}$  denote the participating modalities, with  $M = |\mathcal{M}|$ , and let  $t_m^p \in \mathbb{R}^d$  be the input token of modality  $m$  at patch  $p$ . We omit the batch dimension for clarity.

Within each corresponding patch, separate mappings learn shared and modality-specific representations with distinct training constraints. Both representations contribute to fusion through residual token updates and the prediction of modality contributions and object context. Object-region supervision and predicted foreground gates give their learning and use a detection-specific focus. The gates and modality weights regulate input-token updates and scaling, while object context modulates attention queries and fused outputs. Local interaction is followed by feature transformation and spatial reassembly to produce fused BEV features for the 3D detection head. Control signals are predicted from input features during both training and inference; ground-truth (GT) boxes provide supervision targets for representation constraints and object-related predictions only during training.

### 3.2 Shared and Modality-Specific Representation Decoupling

Spatially corresponding multimodal features contain both responses shared across sensors and cues specific to each modality. To explicitly organize these two components, we introduce two independent mappings for each modality:

$$c_m^p = \phi_m^c(t_m^p), \quad u_m^p = \phi_m^u(t_m^p), \quad c_m^p, u_m^p \in \mathbb{R}^d. \quad (1)$$

Here,  $c_m^p$  and  $u_m^p$  denote shared and modality-specific representations, respectively, and  $\phi_m^c$  and  $\phi_m^u$  are lightweight multilayer perceptrons. Their parameters are separate across modalities; correspondence between shared representations is established through training constraints.

To connect representation learning to detection, we construct a foreground patch set  $\mathcal{P}_{\text{fg}}$  for supervision from training annotations. A patch is marked as foreground when its center lies within the BEV footprint of a ground-truth 3D box expanded by a predefined margin. The representation mappings in Equation (1) operate at all spatial locations; this set only specifies where the following three training constraints are evaluated:

$$\begin{aligned} \mathcal{L}_c &= \mathbb{E}_{p \in \mathcal{P}_{\text{fg}}, m < n} [1 - \cos(c_m^p, c_n^p)], \\ \mathcal{L}_u &= \mathbb{E}_{p \in \mathcal{P}_{\text{fg}}, m < n} [\cos(u_m^p, u_n^p) - \delta]_+, \\ \mathcal{L}_{\text{sep}} &= \mathbb{E}_{p \in \mathcal{P}_{\text{fg}}, m} [|\cos(c_m^p, u_m^p)|]. \end{aligned} \quad (2)$$

Here,  $\cos(\cdot, \cdot)$  denotes cosine similarity,  $[a]_+ = \max(a, 0)$ , and  $\delta$  is a similarity margin. Each expectation denotes an arithmetic mean over the indicated foreground patches and modalities or unordered modality pairs.  $\mathcal{L}_c$  encourages agreement between shared representations of the same region;  $\mathcal{L}_u$  penalizes excessive cross-modal similarity between modality-specific representations; and  $\mathcal{L}_{\text{sep}}$  limits similarity between the two representations of each modality. These soft constraints establish distinct roles for the two branches within object regions, alongside the detection objective. The foreground set is used only for training supervision; representations at all spatial locations are computed directly from input tokens.

### 3.3 Representation-Driven Fusion

We use the decoupled representations to determine how local information participates in fusion. For patch  $p$ , let  $\bar{c}^p = M^{-1} \sum_m c_m^p$  and  $\bar{u}_{\text{abs}}^p = M^{-1} \sum_m |u_m^p|$  denote the mean shared representation and the mean magnitude of modality-specific representations. Their concatenation forms a joint descriptor  $h^p = [\bar{c}^p; \bar{u}_{\text{abs}}^p]$ , where absolute values are elementwise.

**Input-token modulation.** The joint descriptor predicts a patch-level foreground gate. Each modality contribution score considers the original token, its two representations, and the deviation of its shared representation from the cross-modal mean:

$$\begin{aligned} g^p &= \sigma(f_g(h^p)), \\ s_m^p &= f_m([t_m^p; c_m^p; u_m^p; |c_m^p - \bar{c}^p|]), \\ \alpha_m^p &= \frac{\exp(s_m^p/\tau)}{\sum_{n \in \mathcal{M}} \exp(s_n^p/\tau)}. \end{aligned} \quad (3)$$

Here,  $\sigma$  is the sigmoid function,  $f_g$  and  $f_m$  are learned prediction heads, and  $\tau > 0$  is a temperature. The gate  $g^p \in (0, 1)$  is a continuous foreground-relevance score predicted directly from multimodal representations and shared across modalities within a patch. This head neither generates bounding boxes nor requires previously detected object regions. Gates are computed at every patch through Equation (3) and directly used for continuous modulation. The weight  $\alpha_m^p$  represents the learned relative contribution of modality  $m$ . We convert these contributions into bounded scaling factors centered on one and update the input tokens:

$$\begin{aligned} w_m^p &= \text{clip}(1 + \eta g^p (M \alpha_m^p - 1), w_{\min}, w_{\max}), \\ \tilde{t}_m^p &= w_m^p [t_m^p + \lambda g^p (c_m^p + u_m^p)]. \end{aligned} \quad (4)$$

The coefficients  $\eta$  and  $\lambda$  control modality scaling and the representation residual, respectively, and the scaling interval contains one. Uniform modality contributions yield  $w_m^p = 1$ . As the gate approaches zero, the representation residual diminishes and the scaling factors approach one, preserving the original token  $t_m^p$ . The gate therefore regulates local update strength without removing background features through a binary mask, while modality scores determine relative feature scaling.

**Object-context-guided interaction.** To guide cross-modal interaction with information related to detection targets, we also predict object context from  $h^p$ . In the single-class primary setting, an objectness prediction is mapped into the token space:

$$r^p = \sigma(f_{\text{obj}}(h^p)), \quad z^p = \tanh(W_z r^p), \quad z^p \in \mathbb{R}^d. \quad (5)$$

Here,  $f_{\text{obj}}$  is a separate objectness prediction head and  $W_z$  is a learned mapping. The foreground gate controls update strength, while  $z^p$  supplies a context vector derived from predicted objectness.

Multi-class settings can instead use foreground class probabilities to form the context, with the corresponding supervision specified in the appendix.

Let  $Q \in \mathbb{R}^{N_q \times d}$  be learned queries reused across patches, and let  $\tilde{T}^p \in \mathbb{R}^{M \times d}$  stack the controlled modality tokens. We incorporate the context into both the queries and attention outputs:

$$\begin{aligned} Q'^p &= Q + \beta g^p \mathbf{1}_{N_q} (z^p)^\top, \\ F^p &= \text{MHA}(Q'^p, \tilde{T}^p, \tilde{T}^p), \\ F'^p &= F^p + \gamma g^p \mathbf{1}_{N_q} (z^p)^\top. \end{aligned} \tag{6}$$

MHA denotes multi-head attention with learned query, key, and value projections. The vector  $\mathbf{1}_{N_q}$  broadcasts the local context over all query positions within the patch, and  $\beta$  and  $\gamma$  control the two residual strengths. The updated tokens  $F'^p$  undergo subsequent feature transformation and spatial reassembly to form the fused BEV representation. Shared and modality-specific representations thereby contribute to detection features through input updates, modality contribution control, and context-guided interaction.

### 3.4 Training Objectives and Inference

In the primary setting, foreground indicators  $y^p = \mathbf{1}[p \in \mathcal{P}_{\text{fg}}]$  supervise both the foreground gate  $g^p$  and objectness prediction  $r^p$ . Their losses,  $\mathcal{L}_g$  and  $\mathcal{L}_{\text{ctx}}$ , use binary cross-entropy with positive-class weighting over foreground and background patches in a training batch. The three representation constraints in Equation (2) are evaluated within the foreground set. The training forward pass also uses the feature-predicted  $g^p$  and  $r^p$ ; labels  $y^p$  enter only the loss computation and do not replace these predictions in fusion.

The overall objective is

$$\begin{aligned} \mathcal{L} &= \mathcal{L}_{\text{det}} + \lambda_{\text{scl}} \mathcal{L}_{\text{scl}} \\ &\quad + \lambda_{\text{aux}} (\lambda_c \mathcal{L}_c + \lambda_u \mathcal{L}_u + \lambda_{\text{sep}} \mathcal{L}_{\text{sep}} + \lambda_g \mathcal{L}_g + \lambda_{\text{ctx}} \mathcal{L}_{\text{ctx}}). \end{aligned} \tag{7}$$

Here,  $\mathcal{L}_{\text{det}}$  includes detection classification, box regression, and orientation losses.  $\mathcal{L}_{\text{scl}}$  is the sensor-combination supervision inherited from ASF: the sum of detection losses on single-modality and two-modality feature combinations (Paek and Kong, 2025). It is enabled in the primary K-Radar setting, and training configurations are reported separately for each dataset. The coefficient  $\lambda_{\text{aux}}$  controls the overall weight of the added supervision, while the remaining coefficients balance its components. The forward modulation is differentiable, allowing detection and auxiliary objectives to jointly optimize the representation branches and control heads.

At inference, the model retains the same gate, modality contribution, and context prediction branches and forms detection features through Equations (4)–(6), without constructing GT labels or computing losses. The foreground gate provides continuous control over local fusion; the subsequent detection head predicts object classes, 3D locations, dimensions, and orientations. Network dimensions, loss weights, sensor-combination training details, and dataset-specific adaptations are provided in the appendix.

[TO COMPLETE: Insert the revised architecture figure; verify bounded modality scaling and the gate/context paths against the method equations.]

**Figure 2:** Overview of ObjDec. Modality-specific BEV features are projected into spatially corresponding patch tokens and mapped to shared and modality-specific representations. Foreground gates and modality contributions predicted from these representations jointly regulate input tokens, while object context modulates attention queries and fused outputs. Feature transformation and spatial reassembly produce BEV features for the 3D detection head. The solid forward path uses feature-predicted control signals during both training and inference. Dashed connections carry GT-derived labels to training losses, not to the gate predictor. Unified projection, local cross-sensor attention, and subsequent feature transformation follow ASF (Paek and Kong, 2025).

## 4 Experiments

### 4.1 Experimental Setup

**Datasets and metrics.** We use K-Radar as the primary benchmark and extend the evaluation to V2X-Radar-V to examine the architecture under different sensor inputs and object categories (Paek et al., 2022; Yang et al., 2025). K-Radar v1 evaluates Sedan within a region of  $[0, 72] \times [-6.4, 6.4] \times [-2, 6]$  m, with 10,065 test frames after filtering. Version v2 uses revised labels, expands the lateral and vertical ranges to  $[-16, 16]$  and  $[-2, 7.6]$  m, and evaluates Sedan and Bus/Truck over 13,727 test frames. We report 3D and bird’s-eye-view average precision, denoted by  $AP_{3D}$  and  $AP_{BEV}$ ; Mean on v2 is the arithmetic mean over the two classes. Main K-Radar results use a confidence prefilter of 0.3. Additional thresholds, complete IoU results, and the evaluator differences between v1 and v2 are detailed in Appendices B and C. On V2X, we use deduplicated validation and test sets containing 1,487 and 1,486 frames. We report Moderate  $AP_{R40}$  at IoU thresholds of 0.7, 0.5, and 0.5 for Vehicle, Pedestrian, and Cyclist, respectively.

**Compared methods and model configurations.** We distinguish published results, results associated with released checkpoints, and local training runs. The v1 experiments use the complete ObjDec architecture described in Section 3. For v2, we evaluate DecControlled Strong, which retains shared and modality-specific representations, foreground gating, and modality contribution control, but omits object context and its query/output modulation. This setting examines decoupled control under wider spatial coverage and an additional category. ASF serves as the direct three-sensor fusion reference, while L4DR provides a LiDAR–radar architectural comparison (Paek and Kong, 2025; Huang et al., 2025). The V2X ASF-style baseline retains unified projection, local attention, and subsequent feature transformation, with the ObjDec representation branches and control paths removed from the common backbone; this adaptation does not use SCL. Sources, modalities, and initialization differences are identified in the tables or captions.

**Training and model selection.** The K-Radar ASF/ObjDec models use pretrained modality encoders with frozen parameters, while BatchNorm statistics follow the original training behavior. We use AdamW (Loshchilov and Hutter, 2019) for training. Local v2 comparisons use the same training set, 11 epochs, and an effective batch size of 2, followed by evaluation of the final checkpoint. ASF and Strong use the same pretrained encoders, whereas L4DR trains its native network from scratch; this comparison therefore matches the training schedule and sample budget. The full v1 model and ablated variants undergo preliminary evaluation on 1,000 samples for checkpoint selection, followed by full-set evaluation; Appendix B documents the available selection procedure and the provenance details still to be completed. V2X uses an 80-epoch budget and an effective batch size of 8, selecting checkpoints by validation mean  $AP_{3D}$  before testing. ASF’s sensor-combination supervision is retained in the primary K-Radar configurations and disabled in the V2X adaptation. Network dimensions, learning-rate schedules, and loss weights are provided in Appendix A.

### 4.2 Detection Performance on K-Radar

**Main results on v1.** Table 1 compares methods under the Sedan narrow-region setting. ObjDec achieves 88.36  $AP_{3D}@0.3$  and 88.10  $AP_{BEV}@0.5$ , improving over the results associated with the released ASF checkpoint at the same confidence threshold by 8.04 and 7.77 percentage points. At the stricter  $AP_{3D}@0.5$ , performance increases from 67.19 to 67.50. The gains in this comparison are concentrated at the lower 3D IoU threshold and in BEV detection. We also report a locally retrained ASF reference: ObjDec improves  $AP_{BEV}@0.5$  by 7.73 points, achieves similar  $AP_{3D}@0.5$ , and is 0.22 points lower in  $AP_{3D}@0.3$ . Appendix C reports confidence sensitivity; with the additional confidence filter set to 0.0, ObjDec and the same official ASF archive achieve 88.06 and 87.34

$AP_{3D}@0.3$ . The reported gains are therefore specific to the checkpoints and evaluation settings shown.

**Extended setting on v2.** Table 2 evaluates wider spatial coverage and two object categories. Under the matched training schedule, DecControlled Strong achieves mean  $AP_{3D}@0.3/@0.5/@0.7$  of 66.91/43.05/11.11, exceeding locally trained ASF by 2.14/3.36/1.56 percentage points. All six aggregate 3D/BEV metrics in the complete results improve numerically. Bus/Truck  $AP_{3D}@0.5$  increases from 27.88 to 33.97, showing gains on a category beyond the primary v1 setting. Strong also exceeds the official ASF archive by 1.48 points in mean  $AP_{3D}@0.5$ . The released L4DR checkpoint remains stronger in mean  $AP_{3D}@0.3/@0.5$ , reflecting differences in architecture and training conditions. These results support the applicability of the decoupled-control variant to the extended setting.

**Conditions and sensor availability.** Appendices C and F report weather breakdowns and missing/corrupted-input analyses with fixed model weights. On v2, Strong improves over locally trained ASF in 24 of the 26 valid weather-class entries at IoU 0.3 and 0.5, with decreases in the remaining two entries. On v1, removing the camera yields 88.06  $AP_{3D}@0.3$ , close to 88.36 with all sensors, whereas removing LiDAR reduces it to 57.62. The model thus retains performance under some input changes while remaining dependent on LiDAR. We treat weather groups and sensor availability as conditional analyses and report the complete combinations and exceptions in the appendix.

### 4.3 Component Ablations and Computational Cost

**Representation supervision and fusion control.** Table 3 examines the constraints used to learn the decoupled representations and the mechanisms through which they guide fusion. Removing decoupling supervision retains the shared and modality-specific branches and all forward control paths, but reduces  $AP_{3D}@0.3/@0.5$  from 88.36/67.50 to 80.23/66.16. This comparison supports the contribution of the representation constraints within the proposed architecture. Fixing the foreground gate to one and removing its supervision yields 80.00/64.80, indicating the value of learned spatial modulation. Disabling modality contribution control produces 79.62/64.38, while removing object context yields 80.14/66.23. Under the reported selection and evaluation procedure, all four changes reduce both metrics, supporting the joint learning and use of decoupled representations for detection. Appendix D provides stricter 3D and additional BEV metrics omitted here, together with implementation details of each ablation.

**Computational cost.** ObjDec increases the parameter count from 78.13M to 79.47M relative to ASF, an increase of approximately 1.7%. In paired measurements on an RTX A6000 with batch size 1, standard eager execution takes 84.00 ms for ASF and 84.69 ms for ObjDec. Timing includes preprocessing, device transfer, encoding, fusion, detection decoding, and NMS, while excluding disk loading and collation. With the same CUDA Graph optimization, the times are 59.30 and 59.57 ms, respectively, indicating a small latency overhead for the added computation paths. The optimized path has undergone sampled numerical checks but has not yet been evaluated for full-test AP; Appendix H separately documents its timing protocol, numerical checks, and runtime limitations.

### 4.4 Representation and Qualitative Analysis

**Shared and modality-specific representations.** We analyze all 10,065 v1 test frames, covering 51 sequences and 642,649 foreground patch locations. Figure 3 shows input, shared, and modality-specific representations under normal and heavy-snow conditions. Each point represents the mean feature over one frame’s GT-defined foreground patches; the PCA bases are fitted using all weather

groups. Shared and modality-specific branches exhibit different distribution structures, with residual modality-related structure also visible in shared features. Separately, cosine similarity is computed at matched foreground locations in the original 256-dimensional space and averaged within frames and weather groups. Across seven weather groups, mean cross-modal cosine ranges from 0.951 to 0.959 for shared representations and from 0.446 to 0.549 for modality-specific representations. These statistics describe feature organization rather than establish semantic identifiability: frame-level mismatch controls also yield high shared cosine before centering. We therefore interpret the visualizations alongside the representation-supervision ablation in Table 3. Appendix E provides all-weather PCA, original-space distributions, and correspondence controls.

**Spatial gating and detection.** Figure 4 relates predicted foreground gates to detection outputs in three real test scenes, showing camera projections, complete LiDAR BEV and gate maps, and matched gate detail windows. Gates are predicted directly from multimodal features; GT is used only for training supervision and post-hoc reference overlays or statistics. The native 0.8 m gate grids share a common color scale without smoothing. Selected examples show locally higher responses near objects, while weak responses and responses outside boxes remain. Across all 10,065 frames, equally weighted frame-level foreground and background means are 0.270 and 0.118. Fixed-weight interventions further show sensitivity to gate placement: replacing gates by their frame-wise means or spatially permuting them reduces detection performance, whereas jointly removing the query/output context increments changes the overall 3D@0.3 and BEV@0.5 scores by less than 0.02 percentage points. These interventions differ from retrained component ablations and are reported in Appendix D.3. Appendix E provides all-weather gate examples and statistics, paired ASF/ObjDec detections, and a localization counterexample.

## 4.5 Architecture Evaluation on V2X-Radar-V

**Purpose and setup.** We evaluate the applicability of ObjDec on another dataset with three object categories and different sensor combinations. Models are trained on V2X-Radar-V, retaining shared and modality-specific representations, foreground gating, modality contribution control, and object-context modulation, with context derived from Vehicle, Pedestrian, and Cyclist categories. Local experiments use 8,391 training frames, 1,487 deduplicated validation frames, and 1,486 deduplicated test frames within  $[0, 102.4) \times [-51.2, 51.2) \times [-5, 3)$  m. All local runs use 0.16 m grids, an 80-epoch budget, and an effective batch size of 8.

**Representative methods and comparison groups.** Table 4(a) provides published reference results for PointPillars, CenterPoint, PV-RCNN, SQDNet, BEVDepth, and RPFA-Net, covering LiDAR, camera, and radar detection (Yang et al., 2025). These results use the original paper’s validation protocol, which differs from our local data version, splits, and evaluation region; no ranking or gain is computed across panels. Table 4(b) focuses on locally trained fusion methods. The C+L+R group compares ASF-style and ObjDec using common encoders and detection heads,  $2 \times 2$  patches, and four queries. The L+R group compares separately trained L4DR and ObjDec-LR with their respective native networks. ASF-style is a local adaptation without SCL.

**Matched-modality detection results.** Table 4(b) reports final test results for checkpoints selected by validation mean 3D AP. With L+R inputs, ObjDec-LR achieves mean 3D and BEV AP of 79.65% and 83.81%, exceeding L4DR by 1.41 and 1.25 percentage points. Vehicle, Pedestrian, and Cyclist 3D AP improve by 0.71, 1.58, and 1.93 points, respectively, with improvements in BEV AP for all three classes. With C+L+R inputs, ObjDec achieves mean 3D and BEV AP of 81.38% and 86.27%, exceeding ASF-style by 0.08 and 0.55 points. The mean 3D difference is small; gains are concentrated in Cyclist and BEV metrics, while Vehicle and Pedestrian 3D AP remain lower by 0.47 and 0.24 points. These matched-modality results support the applicability of ObjDec to another

**Table 1: Main K-Radar v1 results.** Detection results for the K-Radar Sedan narrow-region setting. C, L, and R denote camera, LiDAR, and 4D radar. The final three rows use the project’s v1 labels and a confidence filter of 0.3. The released ASF result is taken from the checkpoint-associated archive, with local retraining reported separately. Other rows retain their published or public-log settings and are not uniformly reproduced under identical labels, preprocessing, and evaluators. Horizontal rules separate published references from project-protocol results.

| Method                          | Sensors | Source                      | AP <sub>3D</sub><br>@0.3 | AP <sub>3D</sub><br>@0.5 | AP <sub>BEV</sub><br>@0.3 | AP <sub>BEV</sub><br>@0.5 |
|---------------------------------|---------|-----------------------------|--------------------------|--------------------------|---------------------------|---------------------------|
| RTNH (Paek et al., 2022)        | R       | Reported                    | 37.40                    | 14.10                    | 41.10                     | 36.00                     |
| RTNH (Paek et al., 2022)        | L       | Reported                    | 72.70                    | 37.80                    | 76.50                     | 66.30                     |
| 3D-LRF (Chae et al., 2024)      | L+R     | Reported                    | 74.80                    | 45.20                    | 84.00                     | 73.60                     |
| L4DR (Huang et al., 2025)       | L+R     | Public log                  | 77.96                    | 53.50                    | 79.49                     | 77.54                     |
| DLRFusion (Chae et al., 2025)   | L+R     | Reported                    | 74.80                    | 45.70                    | 82.90                     | 73.20                     |
| RAF on L4DR (Park et al., 2026) | C+L+R   | Reported                    | —                        | 57.40                    | —                         | 82.00                     |
| AW-MoE (Lin et al., 2026)       | L+R     | Reported                    | 83.90                    | 61.50                    | 88.20                     | 84.20                     |
| AW-MoE-LRC (Lin et al., 2026)   | C+L+R   | Reported                    | 84.30                    | 61.80                    | —                         | —                         |
| ASF (Paek and Kong, 2025)       | C+L+R   | Released checkpoint archive | 80.31                    | 67.19                    | 80.78                     | 80.33                     |
| ASF (Paek and Kong, 2025)       | C+L+R   | Local retraining            | 88.57                    | 67.49                    | 89.01                     | 80.36                     |
| ObjDec                          | C+L+R   | Ours                        | 88.36                    | 67.50                    | 88.84                     | 88.10                     |

dataset and different input combinations. The L+R comparison retains native network differences, so its gains cannot be attributed solely to an individual fusion component. Appendices G.1–G.3 retain the Concat control on the common backbone, class-wise BEV scores, spatial-granularity analyses, and best/last-checkpoint results. Appendix G.4 provides the supplementary VoD experiments.

**Table 2: Extended K-Radar v2 evaluation.** Results on the two-class K-Radar v2 setting. Mean averages Sedan and Bus/Truck equally. All rows use the aligned v2.0 labels, revised evaluator, and confidence filter of 0.3. Released-model references are grouped separately from 11-epoch local runs. ASF and Strong share pretrained encoders, whereas local L4DR is trained from scratch with different modalities, architecture, and pretraining cost. Strong is a decoupled-control variant without object context. Complete class-wise and BEV results are provided in Appendix C.

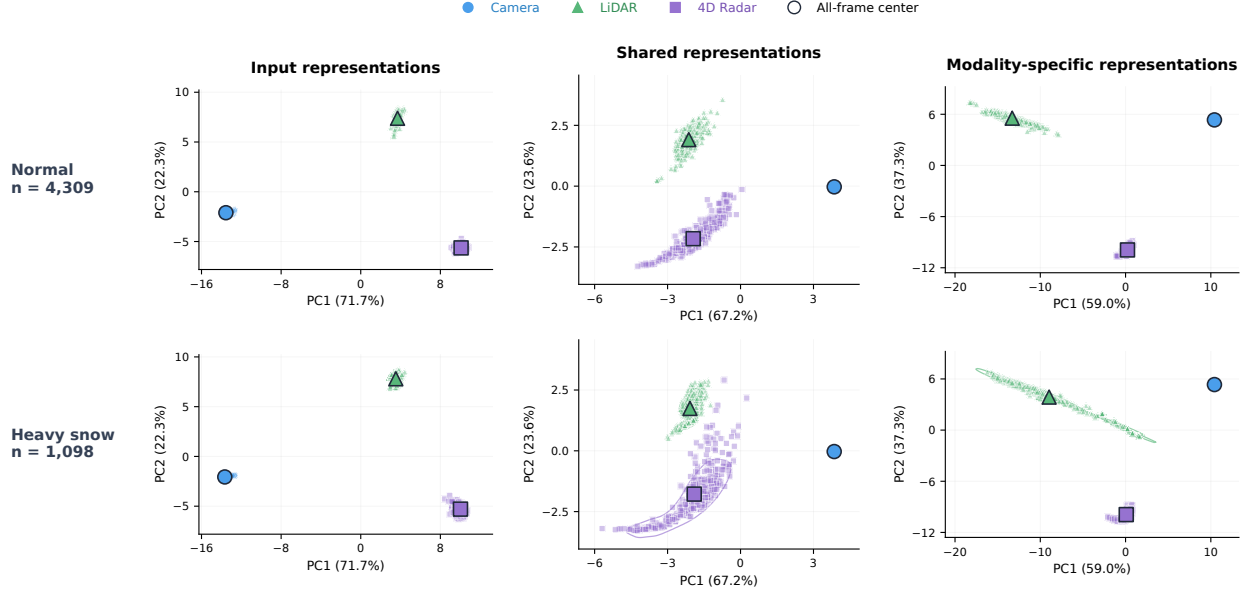
| Method               | Sensors | Training / source                          | Mean<br>AP <sub>3D</sub><br>@0.3 | Mean<br>AP <sub>3D</sub><br>@0.5 | Mean<br>AP <sub>3D</sub><br>@0.7 | Mean<br>AP <sub>BEV</sub><br>@0.5 |
|----------------------|---------|--------------------------------------------|----------------------------------|----------------------------------|----------------------------------|-----------------------------------|
| ASF                  | C+L+R   | Released-result archive                    | 66.55                            | 41.57                            | 9.89                             | 62.45                             |
| L4DR                 | L+R     | Released checkpoint,<br>aligned evaluation | 69.48                            | 48.23                            | 10.86                            | 65.23                             |
| ASF                  | C+L+R   | Local, 11 epochs                           | 64.77                            | 39.70                            | 9.55                             | 58.77                             |
| L4DR                 | L+R     | Local, 11 epochs                           | 65.53                            | 43.84                            | 9.82                             | 60.73                             |
| DecControlled Strong | C+L+R   | Ours, 11 epochs                            | 66.91                            | 43.05                            | 11.11                            | 60.93                             |

**Table 3: Component ablations.** Component ablations on K-Radar v1 with C+L+R and a confidence filter of 0.3. Each variant undergoes checkpoint selection through a preliminary 1,000-sample evaluation before full-set evaluation. Removing decoupling supervision retains the shared and modality-specific branches. Removing object context disables its supervision and both query/output injections. Removing foreground gating fixes the gate to one and disables its supervision. Selection and implementation details are provided in Appendices B and D [**TO COMPLETE: selection records**].

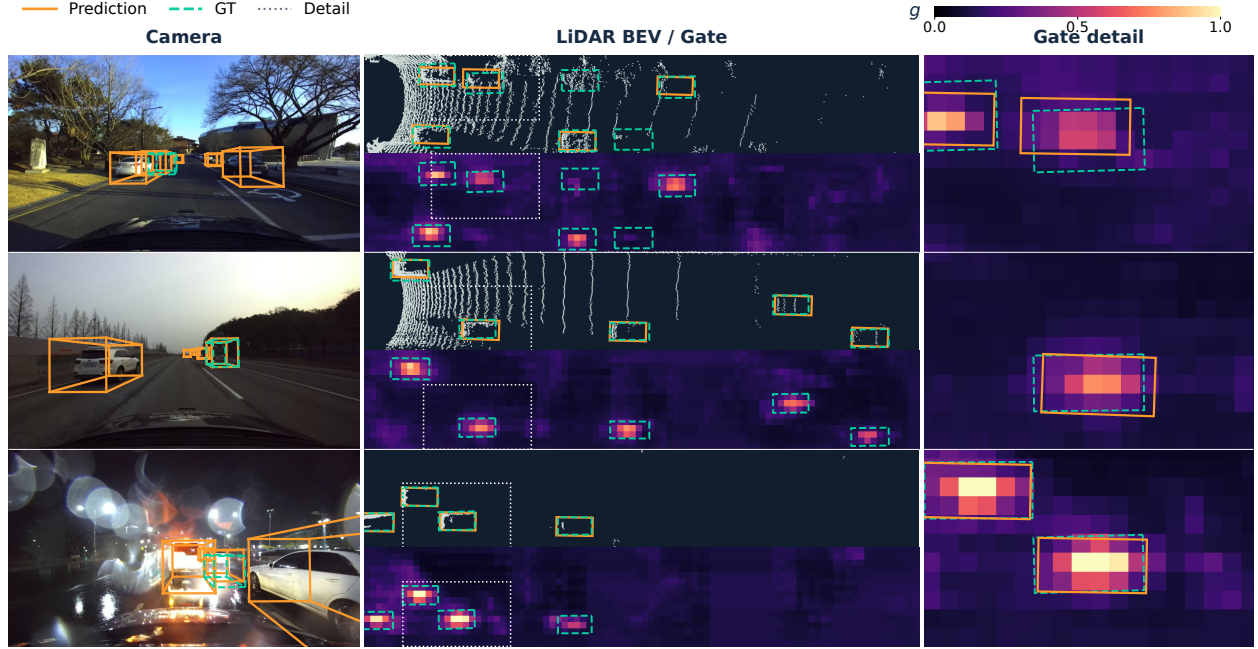
| Variant                           | AP <sub>3D</sub><br>@0.3 | AP <sub>3D</sub><br>@0.5 | AP <sub>BEV</sub><br>@0.5 |
|-----------------------------------|--------------------------|--------------------------|---------------------------|
| Full ObjDec                       | <b>88.36</b>             | <b>67.50</b>             | <b>88.10</b>              |
| w/o modality contribution control | 79.62                    | 64.38                    | 79.91                     |
| w/o object context                | 80.14                    | 66.23                    | 79.96                     |
| w/o decoupling supervision        | 80.23                    | 66.16                    | 80.04                     |
| w/o foreground gating             | 80.00                    | 64.80                    | 80.12                     |

**Table 4: Published references and local fusion comparisons.** Both panels use Moderate difficulty and strict class IoUs of 0.7/0.5/0.5; Mean is the arithmetic class average. Panel (a) reproduces selected results from Table 4 of the V2X-Radar paper (Yang et al., 2025), reported by the dataset authors without local reproduction. Method citations identify the original architectures; all scores in panel (a) come from the dataset paper. Its means are calculated from published class scores; slash-separated entries in the source denote strict/loose IoUs, not 3D/BEV. Panel (b) reports local  $AP_{R40}$ , comparing three-sensor fusion with common encoders and detection heads, and separately trained L+R architectures. ASF-style omits SCL. Data versions, splits, reporting sets, and ROIs differ across panels, precluding a combined ranking or cross-panel gains. Panel (b) uses validation-selected epochs 80, 70, 80, and 75, respectively. The Concat control and complete BEV/best/last results are retained in Tables G.1 and G.2.

| (a) Published validation references (different protocol) |         |            |               |            |         |          |
|----------------------------------------------------------|---------|------------|---------------|------------|---------|----------|
| Method                                                   | Sensors | Vehicle 3D | Pedestrian 3D | Cyclist 3D | Mean 3D |          |
| PointPillars (Lang et al., 2019)                         | L       | 68.80      | 38.16         | 65.24      | 57.40   |          |
| CenterPoint (Yin et al., 2021)                           | L       | 72.19      | 50.59         | 75.26      | 66.01   |          |
| PV-RCNN (Shi et al., 2020)                               | L       | 79.38      | 58.83         | 78.01      | 72.07   |          |
| SQDNet (Mo et al., 2024)                                 | L       | 79.65      | 58.79         | 79.46      | 72.63   |          |
| BEVDepth (Li et al., 2023)                               | C       | 15.47      | 8.51          | 9.46       | 11.15   |          |
| RPFA-Net (Xu et al., 2021)                               | R       | 30.44      | 10.37         | 11.98      | 17.60   |          |
| (b) Local deduplicated test; validation-selected weights |         |            |               |            |         |          |
| Method                                                   | Sensors | Vehicle 3D | Pedestrian 3D | Cyclist 3D | Mean 3D | Mean BEV |
| ASF-style                                                | C+L+R   | 84.54      | 76.17         | 83.18      | 81.30   | 85.72    |
| ObjDec                                                   | C+L+R   | 84.06      | 75.94         | 84.13      | 81.38   | 86.27    |
| L4DR                                                     | L+R     | 83.17      | 73.25         | 78.31      | 78.24   | 82.56    |
| ObjDec-LR                                                | L+R     | 83.87      | 74.84         | 80.24      | 79.65   | 83.81    |



**Figure 3:** PCA visualization of input, shared, and modality-specific representations on K-Radar v1 under normal and heavy-snow conditions. Each small marker represents the 256-dimensional feature vector averaged over all GT-defined foreground patches of one frame and one modality, projected onto two principal components. Blue circles, green triangles, and purple squares denote camera, LiDAR, and 4D radar, respectively. We display the same 400 randomly sampled frames per weather condition across all columns and modalities. Large outlined markers and density contours are computed using all frames of the corresponding weather group ( $n = 4,309$  and  $1,098$ ); contours enclose approximately 80% of the estimated density where estimation is non-degenerate. A separate PCA basis is fitted for each representation using the full test set with equal total weights for weather groups and modalities; both weather rows share that basis and axis limits within each column. Axis percentages indicate explained variance. GT is used only to select regions for this analysis.



**Figure 4:** Spatial gating and detection in matched scenes. Each row shows the camera image with ObjDec predictions (left), the full LiDAR BEV and predicted foreground gate (middle, top and bottom), and a magnified gate region (right). Orange solid and green dashed boxes denote predictions with scores above 0.3 and GT references, respectively. Camera panels show the selected target’s GT; the BEV and gate panels retain all GT references within their displayed regions. White dotted rectangles mark detail windows. Full BEV views cover  $x \in [0, 72]$  m and  $y \in [-6.4, 6.4]$  m; right-hand windows span  $14 \times 8.4$  m, with forward  $x$  horizontal and lateral  $y$  vertical. All gates use the native 0.8 m grid and a common  $[0, 1]$  scale without smoothing. Foreground/background gate means are 0.213/0.128, 0.259/0.115, and 0.288/0.118 from top to bottom. Gates are predicted directly from features; GT overlays and region statistics are added after prediction. Gate values modulate fusion and are distinct from detection confidence.

## 5 Conclusion

We presented ObjDec, a multi-sensor 3D detection architecture centered on learning and using shared and modality-specific representations in corresponding local regions. Object-region supervision gives representation decoupling a detection-specific focus, while the learned representations contribute directly to token updates and guide cross-modal interaction through foreground gating, modality contribution control, and object context. Experiments on K-Radar, together with component ablations and representation analyses, support the effectiveness of this design under the reported evaluation settings. Training and evaluation on V2X-Radar-V further demonstrate its applicability to different object categories and sensor combinations. These findings support representation decoupling as a useful organizing principle for local sensor fusion, with object guidance connecting the learned representations to detection.

## References

- Xuyang Bai, Zeyu Hu, Xinge Zhu, Qingqiu Huang, Yilun Chen, Hongbo Fu, and Chiew-Lan Tai. TransFusion: Robust LiDAR-Camera fusion for 3D object detection with transformers. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022. URL <https://arxiv.org/abs/2203.11496>.
- Konstantinos Bousmalis, George Trigeorgis, Nathan Silberman, Dilip Krishnan, and Dumitru Erhan. Domain separation networks. In *Advances in Neural Information Processing Systems*, volume 29, 2016. URL [https://papers.nips.cc/paper\\_files/paper/2016/hash/45fbc6d3e05ebd93369ce542e8f2322d-Abstract.html](https://papers.nips.cc/paper_files/paper/2016/hash/45fbc6d3e05ebd93369ce542e8f2322d-Abstract.html).
- Yujeong Chae, Hyeonseong Kim, and Kuk-Jin Yoon. Towards robust 3D object detection with LiDAR and 4D radar fusion in various weather conditions. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 15162–15172, 2024. URL [https://openaccess.thecvf.com/content/CVPR2024/html/Chae\\_Towards\\_Robust\\_3D\\_Object\\_Detection\\_with\\_LiDAR\\_and\\_4D\\_Radar\\_CVPR\\_2024\\_paper.html](https://openaccess.thecvf.com/content/CVPR2024/html/Chae_Towards_Robust_3D_Object_Detection_with_LiDAR_and_4D_Radar_CVPR_2024_paper.html).
- Yujeong Chae, Heejun Park, Hyeonseong Kim, and Kuk-Jin Yoon. Doppler-aware LiDAR-RADAR fusion for weather-robust 3D detection. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 27197–27208, 2025. URL [https://openaccess.thecvf.com/content/ICCV2025/html/Chae\\_Doppler-Aware\\_LiDAR-RADAR\\_Fusion\\_for\\_Weather-Robust\\_3D\\_Detection\\_ICCV\\_2025\\_paper.html](https://openaccess.thecvf.com/content/ICCV2025/html/Chae_Doppler-Aware_LiDAR-RADAR_Fusion_for_Weather-Robust_3D_Detection_ICCV_2025_paper.html).
- Devamanyu Hazarika, Roger Zimmermann, and Soujanya Poria. MISA: Modality-invariant and -specific representations for multimodal sentiment analysis. In *Proceedings of the ACM International Conference on Multimedia*, 2020. URL <https://arxiv.org/abs/2005.03545>.
- Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2016. URL <https://arxiv.org/abs/1512.03385>.
- Xun Huang, Ziyu Xu, Hai Wu, Jinlong Wang, Qiming Xia, Yan Xia, Jonathan Li, Kyle Gao, Chenglu Wen, and Cheng Wang. L4DR: LiDAR-4DRadar fusion for weather-robust 3D object detection. *Proceedings of the AAAI Conference on Artificial Intelligence*, 39(4):3806–3814, 2025. doi: 10.1609/aaai.v39i4.32397. URL <https://ojs.aaai.org/index.php/AAAI/article/view/32397>.

- Alex H. Lang, Sourabh Vora, Holger Caesar, Lubing Zhou, Jiong Yang, and Oscar Beijbom. PointPillars: Fast encoders for object detection from point clouds. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2019. URL <https://arxiv.org/abs/1812.05784>.
- Hongsheng Li, Lingfeng Zhang, Zexian Yang, Liang Li, Rong Yin, Xiaoshuai Hao, and Wenbo Ding. Weather-conditioned branch routing for robust LiDAR-Radar 3D object detection, 2026. URL <https://arxiv.org/abs/2604.05405>.
- Yinhao Li, Zheng Ge, Guanyi Yu, Jinrong Yang, Zengran Wang, Yukang Shi, Jianjian Sun, and Zeming Li. BEVDepth: Acquisition of reliable depth for multi-view 3D object detection. *Proceedings of the AAAI Conference on Artificial Intelligence*, 37(2):1477–1485, 2023. doi: 10.1609/aaai.v37i2.25233. URL <https://ojs.aaai.org/index.php/AAAI/article/view/25233>.
- Paul Pu Liang, Zihao Deng, Martin Q. Ma, James Y. Zou, Louis-Philippe Morency, and Ruslan Salakhutdinov. Factorized contrastive learning: Going beyond multi-view redundancy. In *Advances in Neural Information Processing Systems*, volume 36, 2023. URL [https://papers.neurips.cc/paper\\_files/paper/2023/hash/6818dcc65fdf3cbd4b05770fb957803e-Abstract-Conference.html](https://papers.neurips.cc/paper_files/paper/2023/hash/6818dcc65fdf3cbd4b05770fb957803e-Abstract-Conference.html).
- Hongwei Lin, Xun Huang, Chenglu Wen, and Cheng Wang. AW-MoE: All-weather mixture of experts for robust multi-modal 3D object detection, 2026. URL <https://arxiv.org/abs/2603.16261>.
- Zhiwei Lin, Zhe Liu, Zhongyu Xia, Xinhao Wang, Yongtao Wang, Shengxiang Qi, Yang Dong, Nan Dong, Le Zhang, and Ce Zhu. RCBEVDet: Radar-camera fusion in bird’s eye view for 3D object detection. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024. URL <https://arxiv.org/abs/2403.16440>.
- Ye Liu, Zihan Ji, and Hongmin Cai. Plug-and-play feature causality decomposition for multimodal representation learning. In *Advances in Neural Information Processing Systems*, volume 38, 2025. URL [https://proceedings.nips.cc/paper\\_files/paper/2025/hash/55123f38c9f4acf417335cff41be6e27-Abstract-Conference.html](https://proceedings.nips.cc/paper_files/paper/2025/hash/55123f38c9f4acf417335cff41be6e27-Abstract-Conference.html).
- Zhijian Liu, Haotian Tang, Alexander Amini, Xinyu Yang, Huizi Mao, Daniela Rus, and Song Han. BEVFusion: Multi-task multi-sensor fusion with unified bird’s-eye view representation. In *IEEE International Conference on Robotics and Automation (ICRA)*, 2023. URL <https://arxiv.org/abs/2205.13542>.
- Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. In *International Conference on Learning Representations*, 2019. URL <https://openreview.net/forum?id=Bkg6RiCqY7>.
- Yujian Mo, Yan Wu, Junqiao Zhao, Zhenjie Hou, Wei-quan Huang, Yinghao Hu, Jijun Wang, and Jun Yan. Sparse query dense: Enhancing 3D object detection with pseudo points. In *Proceedings of the 32nd ACM International Conference on Multimedia*, pages 409–418, 2024. doi: 10.1145/3664647.3681420. URL <https://doi.org/10.1145/3664647.3681420>.
- Batyrbek Mukhatbekov, Dong-Hee Paek, Woo-Jin Jung, and Seung-Hyun Kong. SRF: Stereo-Radar Fusion for 3D Object Detection in Adverse Weather Conditions. In *IEEE Intelligent Vehicles Symposium*, 2026. URL <https://ave.kaist.ac.kr/2026/02/12/srf-stereo-radar-fusion-for-3d-object-detection-in-adverse-weather-conditions/>.

- Dong-Hee Paek and Seung-Hyun Kong. Availability-aware sensor fusion via unified canonical space. In *Advances in Neural Information Processing Systems*, volume 38, 2025. URL [https://proceedings.neurips.cc/paper\\_files/paper/2025/hash/80bd5c815cdb033ac23eb27605adaaba-Abstract-Conference.html](https://proceedings.neurips.cc/paper_files/paper/2025/hash/80bd5c815cdb033ac23eb27605adaaba-Abstract-Conference.html).
- Dong-Hee Paek, Seung-Hyun Kong, and Kevin Tirta Wijaya. K-Radar: 4D radar object detection for autonomous driving in various weather conditions. In *Advances in Neural Information Processing Systems*, volume 35, 2022. URL <https://arxiv.org/abs/2206.08171>.
- Andras Palffy, Ewoud Pool, Srimannarayana Baratam, Julian F. P. Kooij, and Darius M. Gavrilă. Multi-class road user detection with 3+1D radar in the View-of-Delft dataset. *IEEE Robotics and Automation Letters*, 7(2):4961–4968, 2022. doi: 10.1109/LRA.2022.3147324. URL <https://repository.tudelft.nl/record/uuid%3A663863c1-35b8-48a5-9bc7-e775df8d7fac>.
- Heejun Park, Jaeseok Jeong, and Kuk-Jin Yoon. RAF: Reliability-aware fusion of camera, LiDAR, and 4D RADAR for robust 3D object detection in adverse weather. In *European Conference on Computer Vision*, 2026. URL <https://arxiv.org/abs/2607.04587>.
- Jaehyun Park, Konyul Park, Daehun Kim, Junseo Park, and Jun Won Choi. Layer-Wise Modality Decomposition for Interpretable Multimodal Sensor Fusion. In *Advances in Neural Information Processing Systems*, 2025a. URL [https://proceedings.neurips.cc/paper\\_files/paper/2025/hash/d3e0aaa57c0d639f32a520cda39aec6d-Abstract-Conference.html](https://proceedings.neurips.cc/paper_files/paper/2025/hash/d3e0aaa57c0d639f32a520cda39aec6d-Abstract-Conference.html).
- Konyul Park, Yecheol Kim, Daehun Kim, and Jun Won Choi. Resilient sensor fusion under adverse sensor failures via multi-modal expert fusion. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 6720–6729, 2025b. URL [https://openaccess.thecvf.com/content/CVPR2025/html/Park\\_Resilient\\_Sensor\\_Fusion\\_Under\\_Adverse\\_Sensor\\_Failures\\_via\\_Multi-Modal\\_Expert\\_CVPR\\_2025\\_paper.html](https://openaccess.thecvf.com/content/CVPR2025/html/Park_Resilient_Sensor_Fusion_Under_Adverse_Sensor_Failures_via_Multi-Modal_Expert_CVPR_2025_paper.html).
- Jonah Philion and Sanja Fidler. Lift, splat, shoot: Encoding images from arbitrary camera rigs by implicitly unprojecting to 3D. In *European Conference on Computer Vision*, 2020. URL <https://arxiv.org/abs/2008.05711>.
- Chengxuan Qian, Shuo Xing, Shawn Li, Yue Zhao, and Zhengzhong Tu. DecAlign: Hierarchical cross-modal alignment for decoupled multimodal representation learning. In *International Conference on Learning Representations*, 2026. URL [https://proceedings.iclr.cc/paper\\_files/paper/2026/file/f7f5f501282771c96bb3fedcc96bedfe-Paper-Conference.pdf](https://proceedings.iclr.cc/paper_files/paper/2026/file/f7f5f501282771c96bb3fedcc96bedfe-Paper-Conference.pdf).
- Shaoshuai Shi, Chaoxu Guo, Li Jiang, Zhe Wang, Jianping Shi, Xiaogang Wang, and Hongsheng Li. PV-RCNN: Point-voxel feature set abstraction for 3D object detection. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 10529–10538, 2020. URL [https://openaccess.thecvf.com/content\\_CVPR\\_2020/html/Shi\\_PV-RCNN\\_Point-Voxel\\_Feature\\_Set\\_Abstraction\\_for\\_3D\\_Object\\_Detection\\_CVPR\\_2020\\_paper.html](https://openaccess.thecvf.com/content_CVPR_2020/html/Shi_PV-RCNN_Point-Voxel_Feature_Set_Abstraction_for_3D_Object_Detection_CVPR_2020_paper.html).
- Sana Tonekaboni, Viktoria Schuster, and Caroline Uhler. MultiLoReFT: Decoupling Shared and Modality-Specific Subspaces in Multimodal Learning via Low-Rank Representation Fine-Tuning, 2026. URL <https://arxiv.org/abs/2607.16789>.
- Keyao Wang, Shuai Liu, Hengda Shi, Lukui Shi, and Haiyong Chen. Beyond duality: A hybrid framework of leveraging shared and private features for RGB-Event object detection. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2026. URL <https://arxiv.org/abs/2607.16789>.

- [//openaccess.thecvf.com/content/CVPR2026/papers/Wang\\_Beyond\\_Duality\\_A\\_Hybrid\\_Framework\\_of\\_Leveraging\\_Shared\\_and\\_Private\\_CVPR\\_2026\\_paper.pdf](https://openaccess.thecvf.com/content/CVPR2026/papers/Wang_Beyond_Duality_A_Hybrid_Framework_of_Leveraging_Shared_and_Private_CVPR_2026_paper.pdf).
- Yi Wang, Conrad M. Albrecht, Nassim Ait Ali Braham, Chenying Liu, Zhitong Xiong, and Xiao Xiang Zhu. Decoupling common and unique representations for multimodal self-supervised learning. In *European Conference on Computer Vision*, 2024. URL [https://www.ecva.net/papers/eccv\\_2024/papers\\_ECCV/papers/04236.pdf](https://www.ecva.net/papers/eccv_2024/papers_ECCV/papers/04236.pdf).
- Yuchen Wu, Kun Wang, Yining Pan, and Na Zhao. CCF: Complementary Collaborative Fusion for Domain Generalized Multi-Modal 3D Object Detection. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2026. URL <https://arxiv.org/abs/2603.23276>.
- Baowei Xu, Xinyu Zhang, Li Wang, Xiaomei Hu, Zhiwei Li, Shuyue Pan, Jun Li, and Yongqiang Deng. RPFA-Net: A 4D RaDAR pillar feature attention network for 3D object detection. In *IEEE International Intelligent Transportation Systems Conference (ITSC)*, pages 3061–3066, 2021. doi: 10.1109/ITSC48978.2021.9564754. URL <https://doi.org/10.1109/ITSC48978.2021.9564754>.
- Junjie Yan, Yingfei Liu, Jianjian Sun, Fan Jia, Shuailin Li, Tiancai Wang, and Xiangyu Zhang. Cross modal transformer: Towards fast and robust 3D object detection. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 18268–18278, 2023. URL [https://openaccess.thecvf.com/content/ICCV2023/html/Yan\\_Cross\\_Modal\\_Transformer\\_Towards\\_Fast\\_and\\_Robust\\_3D\\_Object\\_Detection\\_ICCV\\_2023\\_paper.html](https://openaccess.thecvf.com/content/ICCV2023/html/Yan_Cross_Modal_Transformer_Towards_Fast_and_Robust_3D_Object_Detection_ICCV_2023_paper.html).
- Lei Yang, Xinyu Zhang, Jun Li, Chen Wang, Jiaqi Ma, Zhiying Song, Tong Zhao, Ziyang Song, Li Wang, Mo Zhou, Yang Shen, Kai Wu, and Chen Lv. V2X-Radar: A multi-modal dataset with 4D radar for cooperative perception. In *Advances in Neural Information Processing Systems*, volume 38, 2025. URL [https://proceedings.nips.cc/paper\\_files/paper/2025/hash/a501f3238029713afd57ce7924667a-Abstract-Datasets\\_and\\_Benchmarks\\_Track.html](https://proceedings.nips.cc/paper_files/paper/2025/hash/a501f3238029713afd57ce7924667a-Abstract-Datasets_and_Benchmarks_Track.html).
- Tianwei Yin, Xingyi Zhou, and Philipp Krähenbühl. Center-based 3D object detection and tracking. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 11784–11793, 2021. URL [https://openaccess.thecvf.com/content/CVPR2021/html/Yin\\_Center-Based\\_3D\\_Object\\_Detection\\_and\\_Tracking\\_CVPR\\_2021\\_paper.html](https://openaccess.thecvf.com/content/CVPR2021/html/Yin_Center-Based_3D_Object_Detection_and_Tracking_CVPR_2021_paper.html).
- Jingtong Yue, Zhiwei Lin, Xin Lin, Xiaoyu Zhou, Xiangtai Li, Lu Qi, Yongtao Wang, and Ming-Hsuan Yang. RobuRCDet: Enhancing robustness of radar-camera fusion in bird’s eye view for 3D object detection. In *International Conference on Learning Representations*, 2025. URL [https://proceedings.iclr.cc/paper\\_files/paper/2025/hash/21dabaacda3edba8bb281da45d7cbc17-Abstract-Conference.html](https://proceedings.iclr.cc/paper_files/paper/2025/hash/21dabaacda3edba8bb281da45d7cbc17-Abstract-Conference.html).

## A Implementation and Training Details

### A.1 Representation dimensions and foreground targets

This section specifies tensor dimensions and supervision targets omitted from Section 3. Network settings are listed in Table A.1. Both the shared and modality-specific branches follow LayerNorm–Linear–GELU–Linear–LayerNorm, with separate parameters for each modality. The foreground and modality-scoring heads use a hidden dimension of 128, while the single-class objectness head uses 160. The final modality-scoring layer is initialized to zero, giving equal initial modality contributions. For v1, the final foreground layer has zero weights and a bias of  $-1.2$ . Context projections have no bias and use an initial weight standard deviation of 0.015.

For a BEV patch center  $p = (p_x, p_y)$  and a ground-truth box  $b = (x_b, y_b, l_b, w_b, \theta_b)$ , we rotate their relative displacement into the box coordinate system and assign the foreground target according to Equation (A.1). The margin  $\epsilon$  extends each local box axis by 0.7 m on v1 and V2X, and by 0.6 m for v2 Strong. Membership is determined by the patch center rather than patch–box area overlap. Binary foreground labels form the union of valid box footprints. For multiclass context, overlapping labels are overwritten in GT iteration order, retaining the last assigned class. Padded boxes and boxes with nonpositive dimensions are ignored.

$$\begin{aligned} a_{pb} &= (p_x - x_b) \cos \theta_b + (p_y - y_b) \sin \theta_b, \\ b_{pb} &= -(p_x - x_b) \sin \theta_b + (p_y - y_b) \cos \theta_b, \\ y^p &= \mathbf{1}[\exists b : |a_{pb}| \leq l_b/2 + \epsilon, \quad |b_{pb}| \leq w_b/2 + \epsilon]. \end{aligned} \tag{A.1}$$

Equation (A.1) constructs training targets only. The representation branches and predicted gate are evaluated at every patch, and fusion uses the predicted values during both training and inference. GT outlines in the visualizations are added after prediction. Training labels, predicted gates, and reference overlays therefore belong to distinct processing steps.

### A.2 Auxiliary-loss normalization and boundary cases

Let  $N_+$  and  $N_-$  denote foreground and background patch counts in the current batch. Foreground gating uses binary cross-entropy with positive weight  $\omega_+$  defined in Equation (A.2), averaged over all batch patches. The upper bound is  $k = 30$  for the gate and  $k = 4$  for single-class objectness on v1. The representation constraints in Section 3, Equation (2), are evaluated only at foreground locations. If a batch has no selected foreground, no valid GT, or fewer than two modalities on this training path, the implementation skips the added auxiliary-loss block while retaining the corresponding detection-training path.

$$\omega_+ = \text{clip}\left(\frac{N_-}{\max(N_+, 1)}, 1, k\right). \tag{A.2}$$

On V2X, object context is obtained from three-class softmax probabilities through a bias-free linear projection and tanh. Classification cross-entropy is evaluated only on foreground patches, with inverse-frequency class weights computed from foreground counts in the current batch and clipped according to the configuration. The separate gate retains foreground/background supervision. Table A.2 lists forward-control coefficients and auxiliary-loss weights; effective loss weights are the products of the outer and inner coefficients.

**Table A.1: Representation and fusion dimensions.** Input channels are ordered C/L/R. V2X-LR omits the camera encoder while retaining the corresponding fusion dimensions. The earlier V2X 4×4 variant uses 16 queries and 128-dimensional tokens, also producing 128 reassembled channels. Both representation branches preserve token dimensionality.

| Item                                    | K-Radar v1 ObjDec   | K-Radar v2 Strong | V2X ObjDec, 2×2    |
|-----------------------------------------|---------------------|-------------------|--------------------|
| Input BEV channels, C/L/R               | 256 / 512 / 768     | 256 / 512 / 768   | 64 / 64 / 64       |
| Common projection channels              | 256                 | 256               | 64                 |
| Token dimension                         | 256                 | 256               | 128                |
| Shared/specific MLP hidden dimension    | 256                 | 256               | 128                |
| Patch size, BEV cells                   | 2×2                 | 2×2               | 2×2                |
| Queries per patch                       | 32                  | 32                | 4                  |
| Attention heads                         | 16                  | 16                | 8                  |
| Reassembled fusion channels             | 2048                | 2048              | 128                |
| Gate / modality-score hidden dimensions | 128 / 128           | 128 / 128         | 128 / 128          |
| Object-context hidden dimension         | 160                 | —                 | 160                |
| Context target and activation           | Objectness, sigmoid | —                 | 3 classes, softmax |
| Query/output context modulation         | Enabled             | —                 | Enabled            |
| Sensor-combination supervision          | Enabled             | Enabled           | Disabled           |

### A.3 Optimization and sensor-combination supervision

K-Radar fusion models use pretrained single-modality encoders. Their parameters are frozen, while BatchNorm statistics follow the original training mode. The configuration uses AdamW, an initial learning rate of  $10^{-3}$ , a minimum learning rate of  $10^{-4}$ , weight decay of 0.01, batch size 2, and the cosine schedule of the original training entry point, with an 11-epoch limit. Checkpoint selection is described in Section B.2. On V2X, the shared backbone initializes its camera ResNet-50 (He et al., 2016) from ImageNet and all other parameters from untrained initialization. All encoders are trained, using AdamW with an initial learning rate of  $10^{-3}$  and weight decay of 0.01. The 80-epoch schedule uses micro-batches of two and four accumulation steps, giving an effective batch size of eight, followed by cosine decay by optimizer update to  $10^{-5}$  after one warmup epoch.

K-Radar uses ASF’s sensor-combination supervision (SCL) (Paek and Kong, 2025). For three-sensor inputs, detection losses from the six one- and two-sensor branches are summed without division by the number of combinations. Subsets are selected from the controlled tokens already computed using the full input and share the modulated query. Gate, modality contribution, and context are not independently recomputed for each subset. The subset branches do not receive the additional context output residual used by the main fusion path. SCL is disabled on V2X. This implementation provides detection supervision on feature combinations and differs from the fixed-weight missing-sensor inference in Section F.

**Table A.2: Control and supervision coefficients.** Coefficients follow the main-text equations. The primary configurations do not anneal control strength. Strong has no context path, rather than a trained path disabled only at inference. V2X foreground-class weights differ from v1 binary positive weights as described in Section A.2.

| Item                                             | v1 ObjDec               | v2 Strong              | V2X ObjDec              |
|--------------------------------------------------|-------------------------|------------------------|-------------------------|
| FG margin $\epsilon$ , m                         | 0.7                     | 0.6                    | 0.7                     |
| Specific similarity margin $\delta$              | 0.1                     | 0.1                    | 0.1                     |
| Modality scaling strength $\eta$                 | 0.75                    | 0.65                   | 0.75                    |
| Residual strength $\lambda$                      | 0.08                    | 0.08                   | 0.08                    |
| Temperature $\tau$                               | 1.0                     | 1.0                    | 1.0                     |
| Modality scale clipping interval                 | [0.4, 2.1]              | [0.45, 2.0]            | [0.4, 2.1]              |
| Gate range                                       | [0, 1]                  | [0, 1]                 | [0, 1]                  |
| Gate initial bias                                | -1.2                    | -1.5                   | -1.2                    |
| Query / output context strength, $\beta/\gamma$  | 0.16 / 0.06             | —                      | 0.16 / 0.06             |
| Outer auxiliary weight, $\lambda_{\text{aux}}$   | 0.12                    | 0.10                   | 0.12                    |
| Inner common / specific / separation weights     | 0.12 / 0.015 / 0.10     | 0.12 / 0.015 / 0.10    | 0.12 / 0.015 / 0.10     |
| Inner gate / context weights                     | 0.15 / 0.30             | 0.12 / —               | 0.15 / 0.30             |
| Effective common / specific / separation weights | 0.0144 / 0.0018 / 0.012 | 0.012 / 0.0015 / 0.010 | 0.0144 / 0.0018 / 0.012 |
| Effective gate / context weights                 | 0.018 / 0.036           | 0.012 / —              | 0.018 / 0.036           |
| Gate positive-weight cap                         | 30                      | 30                     | 30                      |
| Objectness / foreground-class weight cap         | 4                       | —                      | 4                       |
| SCL coefficient                                  | 1.0                     | 1.0                    | 0                       |

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**Algorithm A.1** Forward computation and training supervision

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**Input:** available sensor observations, model variant, and optional training GT.

1. Encode  $X$  and project BEV features into spatially corresponding patch tokens  $t$ .
2. Compute shared  $c$  and modality-specific  $u$  for every available patch and modality.
3. Predict foreground gate  $g$  and modality contributions from tokens and their shared/specific representations.
4. If the variant uses object context, predict context  $z$  from its context head.
5. Construct controlled tokens and queries using the main-text forward equations.
6. Apply local cross-modal attention, the enabled context output residual, and PFT.
7. Reassemble the BEV feature and obtain predictions from the detection head.
8. If training:
9. Compute the detector’s training losses using GT.
10. If SCL is enabled, add detection losses on the six controlled-token subsets.
11. Construct foreground/context targets from valid GT boxes using Equation (A.1).
12. If valid foreground exists and at least two modalities are available:
13. Add foreground representation losses, gate loss, and enabled context loss.
14. Return predictions and the total loss with Table A.2 coefficients.
15. Return predictions.

The pseudocode specifies data dependencies rather than low-level operator scheduling. Steps 1–7 are shared by training and inference; GT enters supervision from step 9 onward. The target foreground set in step 11 does not replace the predicted gate in step 3. Step 10 follows the SCL implementation in Section A.3 and is skipped on V2X. Losses follow main-text Equation (7).

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## B Evaluation Protocols and Model Selection

### B.1 AP computation and postprocessing

Additional evaluation settings are listed in Table B.1. K-Radar v1 uses the legacy AP aggregation, selecting every fourth value from 41 precision samples, for 11 values in total. The revised v2 implementation averages all 41 values and uses its corresponding height-center convention. Both are distinguished from the standard AP\_R40 aggregation used on V2X. K-Radar’s reported confidence threshold is an additional prediction-export filter applied after the detection head’s score prefilter. Thus, confidence 0.0 does not remove the head-level prefilter.

For aligned v2 evaluation of L4DR, export, GT, confidence, and the effective rotated-NMS threshold are matched to Strong, with NMS set to 0.01. This differs from L4DR’s native effective configuration. The resulting row therefore includes a postprocessing adaptation rather than only a change to AP aggregation. Table B.2 cross-checks AP sampling and height-center conventions with predictions fixed, illustrating why identical class names and IoU thresholds alone do not establish comparability.

### B.2 Result provenance and checkpoint selection

The K-Radar tables distinguish archived results associated with official weights, local evaluation of official weights, and local training. The ASF archive corresponds to the reported configuration and evaluation settings, but is not described as a simultaneous frame-by-frame reevaluation with the other models. Locally trained ASF and Strong on v2 use the same pretrained encoders; local L4DR trains its native L+R network from scratch. Training data, 11 epochs, and effective batch size are matched, while pretraining cost, sensor inputs, and architecture differ.

The full v1 model and ablated variants undergo checkpoint selection using a 1,000-sample preliminary evaluation before full-set evaluation. **[TO COMPLETE: subset origin and IDs or construction rule, candidate ranges, selection metric, and tie handling.]** Until these records are recovered, we do not label this subset an independent validation set. The local v2 training comparison uses final-epoch weights. Strong is identified as an existing decoupled-control variant and does not validate the complete object-context path.

On V2X, the deduplicated validation set is evaluated at epoch 1 and every fifth epoch thereafter. The best checkpoint is selected by mean Moderate 3D AP at the strict class-specific IoUs, retaining the earlier epoch in a tie. After training, best and epoch-80 last checkpoints are evaluated on both raw and deduplicated validation and test sets. Test performance is not used for selection. Small numerical differences can occur between training-time validation and final independent reevaluation; their provenance is retained, and reevaluated scores are not substituted into training curves.

**Table B.1: Additional protocol details.** Sharing a dataset column does not imply matched initialization or training cost; provenance is specified in Section B.2. The v1 1,000-sample set is identified only by its known selection role, pending confirmation of its split origin. Spatial ranges clarify protocol differences and do not replace essential main-table disclosures.

| Setting                                    | K-Radar v1                          | K-Radar v2                           | V2X-Radar-V local adaptation               |
|--------------------------------------------|-------------------------------------|--------------------------------------|--------------------------------------------|
| Evaluated classes                          | Sedan                               | Sedan; Bus/Truck                     | Vehicle; Pedestrian; Cyclist               |
| Test frames                                | 10,065                              | 13,727                               | 1,486 deduplicated; 1,501 raw              |
| Validation frames                          | Selection subset: B.2               | Final-epoch local comparison         | 1,487 deduplicated; 1,498 raw              |
| x / y / z range, m                         | [0,72] / [-6.4,6.4] / [-2,6]        | [0,72] / [-16,16] / [-2,7.6]         | [0,102.4] / [-51.2,51.2] / [-5,3)          |
| Reported difficulty                        | Project legacy class evaluation     | Project revised class evaluation     | Moderate                                   |
| Primary class IoUs                         | 0.3 / 0.5 / 0.7 for Sedan           | 0.3 / 0.5 / 0.7 for both classes     | 0.7 / 0.5 / 0.5 by class                   |
| Additional loose IoUs                      | —                                   | —                                    | 0.5 / 0.25 / 0.25 by class, in raw reports |
| AP aggregation                             | 11 values from 41 precision samples | Mean of 41 precision values          | AP_R40                                     |
| 3D evaluator z-center parameter            | 1.0                                 | 0.5                                  | Native V2X evaluation convention           |
| Head score prefilter                       | 0.1                                 | 0.1                                  | 0.001                                      |
| Additional export confidence filter        | 0.3; 0.0 supplement                 | 0.3; 0.0 for availability supplement | No K-Radar-style additional 0.3 filter     |
| Effective rotated-NMS IoU                  | 0.01                                | 0.01 for aligned comparison          | 0.1                                        |
| Max detections in local V2X postprocessing | —                                   | —                                    | 500 per class                              |

**Table B.2: v2 evaluator cross-check with fixed predictions.** Two-class means at confidence 0.3. GT, predictions, NMS, and confidence are fixed within each checkpoint; only precision aggregation and the 3D IoU height-center convention change. The z-center parameter does not affect geometric BEV IoU, whereas AP sampling still affects BEV AP. This check does not account for every difference between published and local results.

| Checkpoint    | Precision values averaged | z-center | Mean 3D@0.3 | Mean 3D@0.5 |
|---------------|---------------------------|----------|-------------|-------------|
| Strong        | 11                        | 1.0      | 62.01       | 39.46       |
| Strong        | 11                        | 0.5      | 63.26       | 43.66       |
| Strong        | 41                        | 1.0      | 63.93       | 40.04       |
| Strong        | 41                        | 0.5      | 66.91       | 43.05       |
| L4DR released | 11                        | 1.0      | 66.44       | 46.37       |
| L4DR released | 11                        | 0.5      | 70.86       | 47.75       |
| L4DR released | 41                        | 1.0      | 67.43       | 45.27       |
| L4DR released | 41                        | 0.5      | 69.48       | 48.23       |

## C Complete Results on K-Radar

### C.1 Confidence thresholds and stricter matching

Table C.1 extends the main comparison to IoU 0.7 and the additional confidence setting. At confidence 0.0, ObjDec and the official ASF archive obtain 88.06 and 87.34 for 3D@0.3, a smaller difference than at confidence 0.3 in the main text; their 3D@0.5 scores are 72.83 and 72.95. Rankings and margins for fixed weights can change with export thresholds and IoU, so operating points are reported separately. ObjDec exceeds official ASF at the stricter 3D@0.7 criterion under both confidence settings, while its BEV@0.7 is slightly lower. These results characterize sensitivity to matching conditions without extending an advantage at one operating point to all metrics.

Table C.2 expands the v2 class means into per-class 3D and BEV metrics. For Bus/Truck BEV@0.7, Strong improves over locally trained ASF from 16.84 to 23.13, but remains below the 29.65 obtained by official L4DR under aligned evaluation. The per-class breakdown separates fusion comparisons under matched training schedules from references with different architectures and initialization.

### C.2 Weather groups and aggregation

Tables C.3 and C.4 provide the complete weather breakdown. Total AP is computed from the precision-recall sequence over the entire set and cannot be reconstructed by averaging weather APs. A dash marks a class-weather group with no corresponding GT. In particular, v2 Fog contains no Bus/Truck objects and is excluded from weather averages or gain/loss counts for that class.

On v1, rainy-weather 3D@0.5 is 74.92, exceeding both ASF sources, whereas sleet yields 57.60, below both. On v2, Strong improves over locally trained ASF in most class-weather entries, with decreases for Sedan in light snow at IoU 0.3 and heavy snow at IoU 0.5. These complete breakdowns support analysis across conditions. Weather labels are not treated as controlled interventions, and weather-specific performance does not replace the main evidence for the representation design.

**Table C.1: Complete v1 metrics and confidence settings.** Each pair uses the same checkpoint and changes the export confidence filter, in addition to the detector prefilter. Released ASF results are checkpoint-associated archives. Main metrics are retained as anchors for the additional IoU and threshold results.

| Method               | conf | 3D@0.3 | 3D@0.5 | 3D@0.7 | BEV@0.3 | BEV@0.5 | BEV@0.7 |
|----------------------|------|--------|--------|--------|---------|---------|---------|
| ASF released archive | 0.3  | 80.31  | 67.19  | 18.85  | 80.78   | 80.33   | 62.85   |
| ASF released archive | 0.0  | 87.34  | 72.95  | 18.85  | 88.59   | 86.97   | 62.85   |
| ASF local            | 0.3  | 88.57  | 67.49  | 18.69  | 89.01   | 80.36   | 61.59   |
| ASF local            | 0.0  | 87.96  | 72.96  | 18.68  | 88.80   | 87.41   | 61.57   |
| ObjDec               | 0.3  | 88.36  | 67.50  | 22.04  | 88.84   | 88.10   | 62.63   |
| ObjDec               | 0.0  | 88.06  | 72.83  | 22.02  | 88.84   | 87.18   | 62.63   |

**Table C.2: Complete class-wise v2 metrics.** Confidence is 0.3. These class-wise results complement the class means in the main text. Strong omits object context. Local runs match the training split, 11 epochs, and effective batch size; initialization and modality differences are specified in Appendix B.

| Method                 | Class     | 3D@0.3 | 3D@0.5 | 3D@0.7 | BEV@0.3 | BEV@0.5 | BEV@0.7 |
|------------------------|-----------|--------|--------|--------|---------|---------|---------|
| ASF released archive   | Sedan     | 74.98  | 52.09  | 11.85  | 77.85   | 71.70   | 43.21   |
| ASF released archive   | Bus/Truck | 58.13  | 31.06  | 7.93   | 67.85   | 53.21   | 20.85   |
| L4DR released, aligned | Sedan     | 76.79  | 54.68  | 11.23  | 79.90   | 73.61   | 49.16   |
| L4DR released, aligned | Bus/Truck | 62.17  | 41.78  | 10.49  | 67.12   | 56.86   | 29.65   |
| ASF local              | Sedan     | 74.45  | 51.52  | 11.58  | 77.37   | 71.11   | 42.23   |
| ASF local              | Bus/Truck | 55.09  | 27.88  | 7.52   | 64.27   | 46.43   | 16.84   |
| L4DR local             | Sedan     | 74.49  | 54.28  | 14.08  | 75.13   | 71.59   | 48.20   |
| L4DR local             | Bus/Truck | 56.58  | 33.39  | 5.56   | 61.46   | 49.86   | 22.50   |
| DecControlled Strong   | Sedan     | 74.58  | 52.14  | 12.31  | 77.32   | 71.27   | 44.26   |
| DecControlled Strong   | Bus/Truck | 59.24  | 33.97  | 9.92   | 65.14   | 50.59   | 23.13   |

**Table C.3: Weather-wise v1 3D AP.** Sedan at confidence 0.3. Total is evaluated over the combined set and is not an average of weather APs.

| Method               | IoU | Total | Normal | Over-cast | Fog   | Rain  | Sleet | Light snow | Heavy snow |
|----------------------|-----|-------|--------|-----------|-------|-------|-------|------------|------------|
| ASF released archive | 0.3 | 80.31 | 79.57  | 89.89     | 90.67 | 80.97 | 80.20 | 80.89      | 71.71      |
| ASF local            | 0.3 | 88.57 | 87.83  | 90.16     | 90.77 | 80.92 | 80.51 | 89.45      | 71.64      |
| ObjDec               | 0.3 | 88.36 | 87.66  | 90.39     | 90.57 | 88.90 | 80.42 | 89.28      | 71.41      |
| ASF released archive | 0.5 | 67.19 | 64.56  | 79.71     | 79.57 | 67.33 | 67.28 | 77.63      | 61.61      |
| ASF local            | 0.5 | 67.49 | 64.70  | 80.10     | 79.59 | 68.31 | 66.82 | 78.46      | 61.67      |
| ObjDec               | 0.5 | 67.50 | 65.98  | 80.26     | 79.71 | 74.92 | 57.60 | 78.33      | 60.51      |

**Table C.4: Weather-wise v2 3D AP.** Confidence is 0.3. A dash indicates no GT objects of the class in that weather group and is excluded from any class-weather comparison.

(a) IoU 0.3

| Method                 | Class     | Total | Normal | Over-cast | Fog   | Rain  | Sleet | Light snow | Heavy snow |
|------------------------|-----------|-------|--------|-----------|-------|-------|-------|------------|------------|
| ASF released archive   | Sedan     | 74.98 | 74.14  | 82.35     | 92.26 | 66.37 | 70.43 | 88.24      | 63.61      |
| L4DR released, aligned | Sedan     | 76.79 | 75.52  | 83.57     | 93.60 | 78.25 | 67.17 | 87.71      | 58.49      |
| ASF local              | Sedan     | 74.45 | 73.45  | 82.15     | 92.09 | 65.80 | 70.81 | 89.77      | 63.36      |
| L4DR local             | Sedan     | 74.49 | 73.12  | 80.06     | 92.33 | 76.27 | 54.55 | 82.74      | 51.99      |
| DecControlled Strong   | Sedan     | 74.58 | 73.51  | 84.30     | 94.46 | 68.11 | 72.52 | 89.56      | 63.57      |
| ASF released archive   | Bus/Truck | 58.13 | 53.28  | 75.33     | —     | 7.89  | 59.55 | 88.26      | 68.20      |
| L4DR released, aligned | Bus/Truck | 62.17 | 56.27  | 87.43     | —     | 2.96  | 74.75 | 80.47      | 63.76      |
| ASF local              | Bus/Truck | 55.09 | 50.39  | 65.31     | —     | 3.59  | 50.78 | 84.75      | 68.02      |
| L4DR local             | Bus/Truck | 56.58 | 54.60  | 82.07     | —     | 7.81  | 59.89 | 89.16      | 54.27      |
| DecControlled Strong   | Bus/Truck | 59.24 | 51.63  | 73.82     | —     | 10.08 | 57.47 | 89.56      | 72.34      |

(b) IoU 0.5

| Method                 | Class     | Total | Normal | Over-cast | Fog   | Rain  | Sleet | Light snow | Heavy snow |
|------------------------|-----------|-------|--------|-----------|-------|-------|-------|------------|------------|
| ASF released archive   | Sedan     | 52.09 | 49.63  | 53.38     | 80.55 | 46.85 | 44.05 | 59.25      | 49.52      |
| L4DR released, aligned | Sedan     | 54.68 | 50.21  | 56.53     | 83.81 | 53.03 | 47.42 | 62.32      | 53.33      |
| ASF local              | Sedan     | 51.52 | 48.90  | 57.16     | 79.89 | 45.96 | 44.83 | 59.25      | 51.24      |
| L4DR local             | Sedan     | 54.28 | 52.33  | 49.24     | 83.44 | 54.95 | 41.22 | 59.28      | 38.72      |
| DecControlled Strong   | Sedan     | 52.14 | 49.39  | 59.62     | 80.30 | 48.30 | 49.13 | 60.75      | 50.49      |
| ASF released archive   | Bus/Truck | 31.06 | 23.93  | 47.04     | —     | 2.48  | 38.83 | 70.81      | 32.55      |
| L4DR released, aligned | Bus/Truck | 41.78 | 36.60  | 76.06     | —     | 0.68  | 62.92 | 53.30      | 41.11      |
| ASF local              | Bus/Truck | 27.88 | 21.16  | 46.68     | —     | 1.94  | 30.29 | 70.58      | 32.28      |
| L4DR local             | Bus/Truck | 33.39 | 32.18  | 55.89     | —     | 6.60  | 43.05 | 69.00      | 17.00      |
| DecControlled Strong   | Bus/Truck | 33.97 | 23.94  | 49.65     | —     | 5.05  | 38.48 | 75.09      | 39.37      |

## D Component Interventions and Additional Metrics

### D.1 What each ablation changes

This section specifies the interventions in main-text Table 3. Disabling modality contribution control sets the scaling strength in Equation (4) to zero, giving unit modality scaling while retaining representation residuals, predicted gating, and context. Disabling object context removes its supervision and its injection into queries and attention outputs. Disabling decoupling supervision sets only the shared-alignment, modality-specific similarity, and inter-branch separation loss weights to zero; both representation branches still generate features and participate in forward control. Disabling the foreground gate fixes its value to one and removes its supervision, retaining the other branches.

These definitions distinguish supervision changes, gating amplitude, and the forward paths themselves. In particular, removing decoupling supervision does not remove the shared/specific networks and is not equivalent to Concat.

### D.2 Strict 3D matching and BEV results

Table D.1 reports 3D@0.7, BEV@0.3, and BEV@0.7, which are omitted from the main ablation table. Disabling modality contribution control reduces 3D@0.7 from 22.04 to 11.08, while BEV@0.7 remains at 61.36, close to the full model’s 62.63. This intervention affects strict 3D matching and horizontal overlap differently; BEV scores alone therefore do not characterize its effect on 3D localization. All four ablations underperform the full model on the listed supplementary metrics. These observations follow the current single-seed experiments and selection procedure and do not establish statistical significance.

### D.3 Fixed-weight spatial-gate and context interventions

Table D.2 complements the retrained ablations with inference-time interventions on the final K-Radar v1 checkpoint. Each setting uses the same encoded features per frame. Replacing the gate by its frame-wise mean preserves its mean amplitude but removes spatial variation; permuting its values preserves the per-frame distribution while changing correspondence with local features. The three fixed seeds are repeated permutations, not independently trained models. GT is not used to generate the modified signals.

Frame-mean gates reduce 3D@0.3/BEV@0.5 from 88.36/88.11 to 53.90/53.16, while the three permutations average 21.29/21.01. This supports the trained model’s dependence on spatial correspondence. The same gate affects token residuals, modality scaling, and gated context; this

**Table D.1: Additional ablation metrics.** v1 at confidence 0.3. The 3D@0.3/@0.5 and BEV@0.5 columns already shown in the main ablation table are omitted. The full-model row follows the same JSON as Table C.1; ablated rows follow their experiment records.

| Variant                           | 3D@0.7 | BEV@0.7 | BEV@0.3 |
|-----------------------------------|--------|---------|---------|
| Full ObjDec                       | 22.04  | 62.63   | 88.84   |
| w/o modality contribution control | 11.08  | 61.36   | 80.38   |
| w/o object context                | 17.10  | 54.91   | 80.68   |
| w/o decoupling supervision        | 16.49  | 55.71   | 80.65   |
| w/o foreground gate               | 19.41  | 55.49   | 80.65   |

**Table D.2:** Fixed-weight inference interventions on the complete K-Radar v1 test set (10,065 frames; C+L+R; confidence filter 0.3; AP11). All settings reuse the same encoded inputs per frame. Only the indicated control signal is changed; three shuffle seeds are inference permutations, not training runs. The paired baseline differs slightly from the archived main-table evaluation (maximum difference 0.0213 percentage points across six metrics), so all intervention comparisons use this baseline. All AP values are percentages.

| Setting                    | 3D AP   |         |         | BEV AP  |         |         |
|----------------------------|---------|---------|---------|---------|---------|---------|
|                            | @0.3    | @0.5    | @0.7    | @0.3    | @0.5    | @0.7    |
| Paired baseline            | 88.3599 | 67.4838 | 22.0603 | 88.8370 | 88.1066 | 62.6532 |
| Frame-mean gate            | 53.9015 | 40.2109 | 7.9593  | 54.2623 | 53.1635 | 37.6025 |
| Shuffled gate (260923)     | 21.3775 | 17.5284 | 4.7356  | 21.5984 | 21.0439 | 16.6446 |
| Shuffled gate (260924)     | 21.2514 | 17.4692 | 6.2176  | 21.5240 | 20.9864 | 16.5998 |
| Shuffled gate (260925)     | 21.2448 | 12.9107 | 3.9612  | 21.4998 | 20.9850 | 16.6030 |
| No query increment         | 88.3492 | 67.4711 | 22.1068 | 88.8342 | 88.0989 | 62.7320 |
| No output increment        | 88.3599 | 67.4833 | 22.0565 | 88.8358 | 88.1066 | 62.6516 |
| No query/output increments | 88.3453 | 67.4742 | 22.1096 | 88.8342 | 88.0980 | 62.7320 |
| Mean of three shuffles     | 21.2913 | 15.9694 | 4.9715  | 21.5407 | 21.0051 | 16.6158 |

experiment does not isolate those paths or attribute the entire decline to representation decoupling. Internal-signal distribution changes also distinguish these results from retraining without a gate.

Jointly removing the query and output context increments changes 3D@0.3 and BEV@0.5 by  $-0.0146$  and  $-0.0087$  percentage points, respectively; all six overall AP changes have magnitudes below 0.08 points. Removing these residual increments retains cross-modal attention and the trained representations. These results do not establish independently large inference contributions for the context paths, nor do they remove their possible training-time effects. The paired baseline is evaluated in the same pass as the interventions and is kept separate from the historical main-table scores.

## E Representation Statistics and Additional Visualizations

### E.1 Statistical units and weighted PCA

For frame  $i$ , modality  $m$ , and location  $p$ , let  $x_{im}^p$  denote a representation and  $\mathcal{P}_i$  the GT foreground locations. Equation (E.1) defines the frame-mean vector  $\mu_{im}$  and the mean matched-location cosine  $s_i^{mn}$ . Weather-group cosines average  $s_i^{mn}$  equally over frames. This differs both from taking the cosine of frame-mean vectors and from pooling all patches with equal patch weights. All 10,065 analyzed frames contain valid foreground, totaling 642,649 foreground spatial locations. Table E.1 separates modality pairs.

$$\mu_{im} = \frac{1}{|\mathcal{P}_i|} \sum_{p \in \mathcal{P}_i} x_{im}^p, \quad s_i^{mn} = \frac{1}{|\mathcal{P}_i|} \sum_{p \in \mathcal{P}_i} \cos(x_{im}^p, x_{in}^p). \quad (\text{E.1})$$

Input, Shared, and Specific each have a separately fitted PCA basis. Within each representation type, all modalities and weather groups share that basis. PCA is fitted to  $\mu_{im}$ , weighting each vector by  $1/(|\mathcal{W}|N_wM)$ , where  $N_w$  is its weather-group frame count and  $M = 3$ . The leading two eigenvectors of the weighted centered covariance are used, without per-vector normalization or channel standardization. Each weather group and modality therefore contributes equally to fitting. Different representation columns have different bases and scales, so absolute Euclidean distances are not compared across columns.

The seven-weather scatter plot displays up to 400 frames per weather group, including all 383 overcast frames. Displayed frame indices are shared across modalities and representation panels;

centroids and density contours use all frames. With the random seed and PCA bases fixed, we do not refit for visual separation, add jitter, or clip outliers. Figure E.1 expands the main PCA visualization to all weather groups.

## E.2 High-dimensional similarity distributions and weather variation

Table E.1 shows that Specific features are not approximately orthogonal for every modality pair: C–R cosines remain around 0.65 across the seven weather groups, while pairs involving LiDAR vary more across groups. This is compatible with the relative organization encouraged by the two branches and does not require modality-specific features to contain no information in common.

Figure E.2 and Table E.2 further show frame-level similarity distributions. The L–R Shared medians are 0.9632 in normal weather and 0.9662 in heavy snow. For Specific, the 5th–95th percentile range changes from 0.2113–0.4948 to 0.2578–0.7363. Violin width represents empirical density; thick and thin intervals show the 25th–75th and 5th–95th percentiles. These describe across-frame variability, not confidence intervals.

As an aggregation-sensitivity check, equal weighting of sequences gives mean cross-modal Specific cosines of approximately 0.440/0.584 in normal/heavy-snow conditions, compared with 0.446/0.537 under equal frame weighting. Shared features retain high consistency, while the precise Specific difference depends on aggregation weights. Weather is associated with sequence, scene, and object composition. High cosine values also cannot exclude a common dominant direction or low variance. These statistics are therefore interpreted together with component ablations, rather than as independent proof of semantic identifiability or a causal weather effect.

**Frame-level correspondence controls.** Figure E.3 compares cosines of foreground-mean vectors for the same frame and for different-sequence frames within the same weather group, using ten fixed permutation seeds. Matched scores use the same retained anchors. Before centering, Shared C–L, C–R, and L–R cosines are 0.979214/0.970201/0.987697 for matched pairs and 0.979205/0.970225/0.987136 for mismatches. After subtracting each modality’s weather-group mean, the matched/mismatched L–R values are 0.148740/−0.014689, while the camera-related matched values remain close to zero. Thus, a common mean direction can account for much of the high uncentered frame-level similarity. This check does not resolve patch-level correspondence because frame averaging discards local information; it also does not isolate the contribution of the Shared branch. It is a control on the interpretation of similarity, rather than proof of either semantic decomposition or branch redundancy.

## E.3 Construction of gate and detection visualizations

Main-text Figure 4 presents three scenes with camera predictions, complete LiDAR BEV and gate maps, and gate detail windows. Foreground/background gate means are 0.213/0.128, 0.259/0.115, and 0.288/0.118 from top to bottom. Full maps cover forward  $x \in [0, 72]$  m and lateral  $y \in [-6.4, 6.4]$  m; detail windows span  $14 \times 8.4$  m. The gate uses the native  $16 \times 90$  array and a common  $[0, 1]$  scale without smoothing. Camera panels show the selected target’s GT, while BEV and gate panels retain all GT references within the displayed region. GT overlays and regional statistics are added after feature-based prediction; no prior detector provides the gate.

Figure E.4 separately retains the seven-weather gate examples selected from 21 complete-gate candidates for scene readability, object arrangement, and local responses. During that export, GT fields were withheld at the fusion-module input and used only afterward. One historical candidate with unexplained repeated-inference differences was excluded. Table E.3 summarizes these seven appendix examples, not the three main-text scenes. Figure E.5 adds the full-test analysis: equally

**Table E.1: Full-set similarity by modality pair.** Cosines in the original 256-dimensional space are averaged over matched foreground locations within each frame, then equally over frames within a weather group.

| Weather    | Frames | Seqs. | FG patches | Shared C-L | Shared C-R | Shared L-R | Specific C-L | Specific C-R | Specific L-R |
|------------|--------|-------|------------|------------|------------|------------|--------------|--------------|--------------|
| Normal     | 4309   | 17    | 317581     | 0.9455     | 0.9584     | 0.9621     | 0.3429       | 0.6545       | 0.3410       |
| Overcast   | 383    | 2     | 33918      | 0.9520     | 0.9574     | 0.9676     | 0.4211       | 0.6501       | 0.4111       |
| Fog        | 1049   | 6     | 43909      | 0.9391     | 0.9505     | 0.9647     | 0.4659       | 0.6612       | 0.4572       |
| Rain       | 1317   | 8     | 125257     | 0.9431     | 0.9565     | 0.9586     | 0.3855       | 0.6511       | 0.3767       |
| Sleet      | 1106   | 7     | 41665      | 0.9584     | 0.9478     | 0.9638     | 0.5112       | 0.6529       | 0.4840       |
| Light snow | 803    | 4     | 36196      | 0.9470     | 0.9582     | 0.9622     | 0.4787       | 0.6590       | 0.4643       |
| Heavy snow | 1098   | 7     | 44123      | 0.9525     | 0.9481     | 0.9601     | 0.4954       | 0.6458       | 0.4693       |

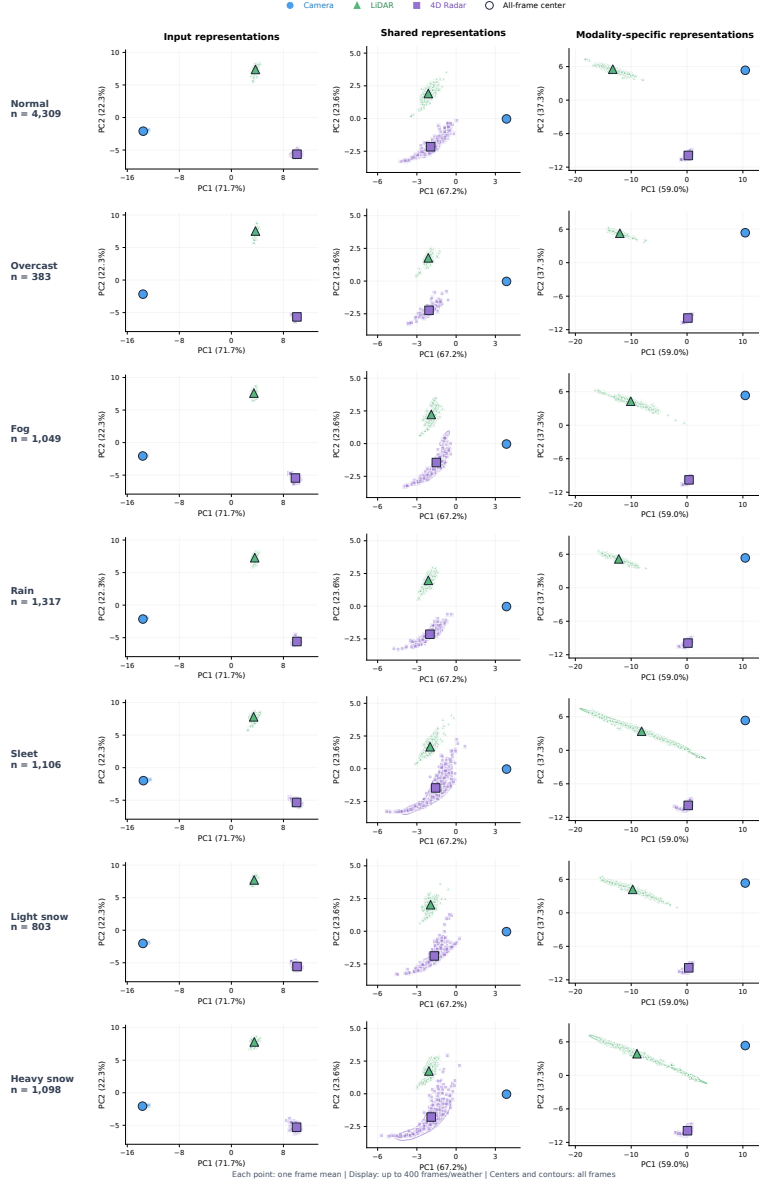
**Table E.2: L-R similarity quantiles.** Each observation is a frame-level mean of matched-patch cosines. Quantiles describe variation across frames, not confidence intervals.

| Weather   | Branch | Pair           | Median | 5th percentile | 95th percentile |
|-----------|--------|----------------|--------|----------------|-----------------|
| normal    | common | LiDAR-4D Radar | 0.9632 | 0.9433         | 0.9776          |
| normal    | unique | LiDAR-4D Radar | 0.3384 | 0.2113         | 0.4948          |
| heavysnow | common | LiDAR-4D Radar | 0.9662 | 0.9141         | 0.9806          |
| heavysnow | unique | LiDAR-4D Radar | 0.4538 | 0.2578         | 0.7363          |

weighted frame-level foreground/background gate means are 0.2701/0.1176, and foreground means exceed background means in 99.06% of frames. Selected examples and full-test statistics have different sampling units and are reported separately.

Figure E.6 compares strictly loaded released ASF and final ObjDec checkpoints on identical test frames, using FP32, batch size 1, the same detection postprocessing, and a score > 0.3 filter. Input-tensor hashes and GT were checked for all 20 candidates in the paired visualization pool. Original camera images, sequence-specific calibration, intrinsics, and distortion are retained. Both methods use identical camera crops, complete BEV ranges, and 12×6 m BEV detail windows. The selected objects’ maximum geometric 3D IoUs among retained predictions change from 0.572/0.000/0.000 to 0.719/0.476/0.741 from top to bottom. These values are neither AP nor the official one-to-one matching outcome; the middle example remains below IoU 0.5.

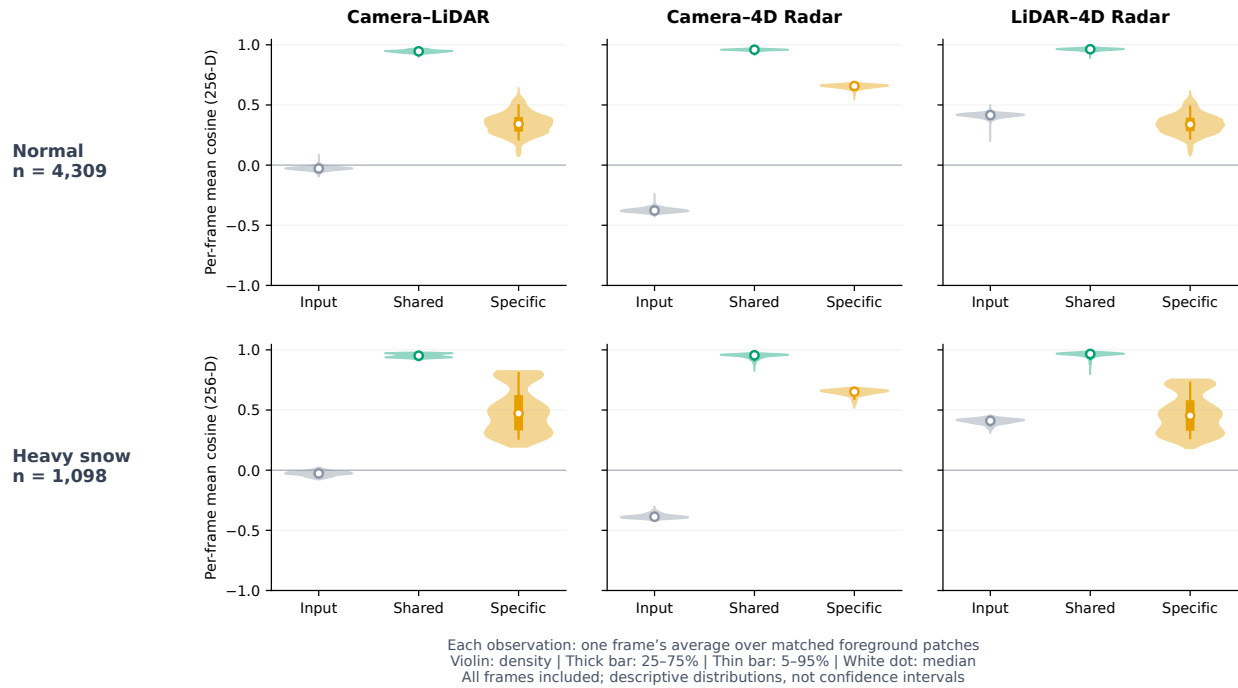
Figure E.7 adds heavy- and light-snow examples: the selected objects’ maximum 3D IoUs change from 0.095 to 0.728 and from 0.354 to 0.223, respectively. The latter also has a locally high gate response, illustrating that foreground-related activation is insufficient to determine final box accuracy. The complete 20-frame candidate atlas is retained as an archival asset; it is not repeated in the manuscript. Selected qualitative cases do not estimate aggregate detection gains.



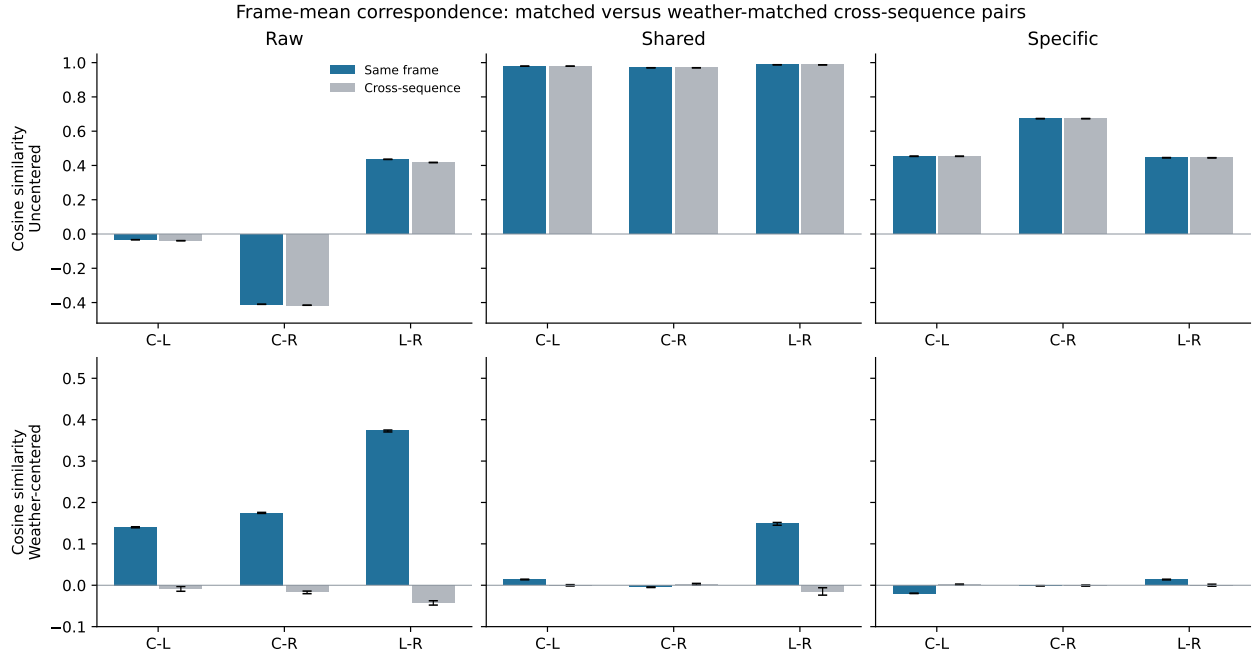
**Figure E.1:** Frame-level representation distributions over the complete K-Radar v1 test set. Rows indicate weather groups; columns show input, shared, and modality-specific representations. Each point is a frame’s mean vector over GT foreground locations, projected using the fixed weighted PCA basis for its column. Blue circles, green triangles, and purple squares denote camera, LiDAR, and 4D radar. Up to 400 frames are displayed per group, including all 383 overcast frames. Large outlined markers and density contours use all frames, and  $n$  denotes the full group size. Each column shares its basis and axis ranges across weather; columns are fitted independently. Residual modality structure in these projections and high-dimensional cosine consistency in Table E.1 characterize different properties.

**Table E.3: Frame-level statistics of the seven appendix gate examples.** Each frame has 1,440 complete patches. Foreground regions use GT footprints expanded by 0.7 m after prediction. Values summarize seven qualitative examples, not weather-wide means.

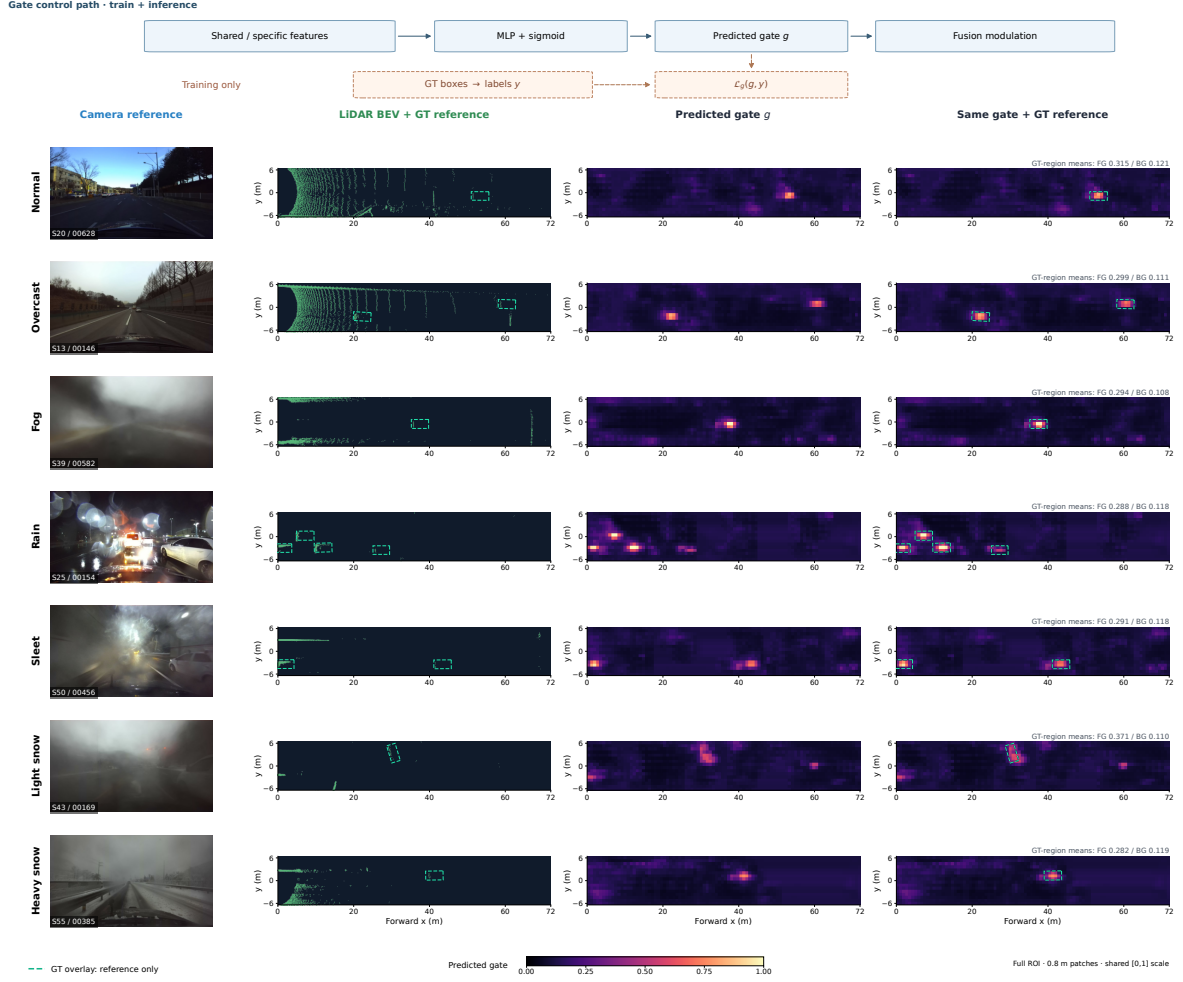
| Weather    | Frame ID       | GT objects | FG patches | FG gate mean | BG gate mean | FG – BG |
|------------|----------------|------------|------------|--------------|--------------|---------|
| Normal     | seq20_rdr00628 | 1          | 28         | 0.3154       | 0.1208       | 0.1946  |
| Overcast   | seq13_rdr00146 | 2          | 60         | 0.2995       | 0.1112       | 0.1883  |
| Fog        | seq39_rdr00582 | 1          | 40         | 0.2942       | 0.1078       | 0.1864  |
| Rain       | seq25_rdr00154 | 4          | 142        | 0.2882       | 0.1181       | 0.1702  |
| Sleet      | seq50_rdr00456 | 2          | 65         | 0.2913       | 0.1179       | 0.1734  |
| Light snow | seq43_rdr00169 | 1          | 35         | 0.3711       | 0.1103       | 0.2608  |
| Heavy snow | seq55_rdr00385 | 1          | 35         | 0.2817       | 0.1186       | 0.1631  |



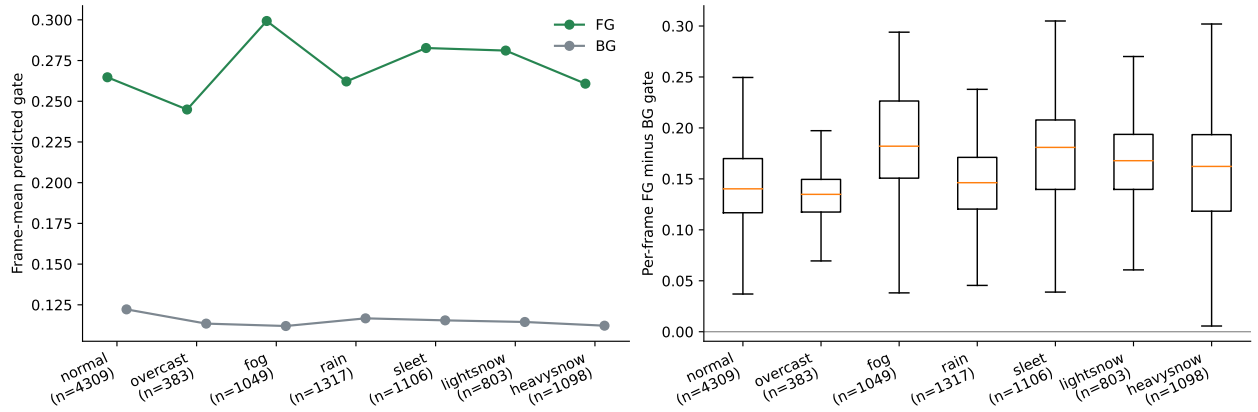
**Figure E.2:** Cosine-similarity distributions in the original 256-dimensional space for three sensor pairs under normal weather and heavy snow. Each observation is the mean matched-foreground-patch cosine within a frame, including all 4,309 and 1,098 frames. Gray, green, and yellow indicate input, shared, and modality-specific representations. Violins show empirical density, white points show medians, and thick/thin intervals span the 25th–75th/5th–95th percentiles, rather than confidence intervals for estimated means. Shared features concentrate at high similarity, while specific-feature distributions vary by modality pair and weather. Similarities are not computed from PCA coordinates.



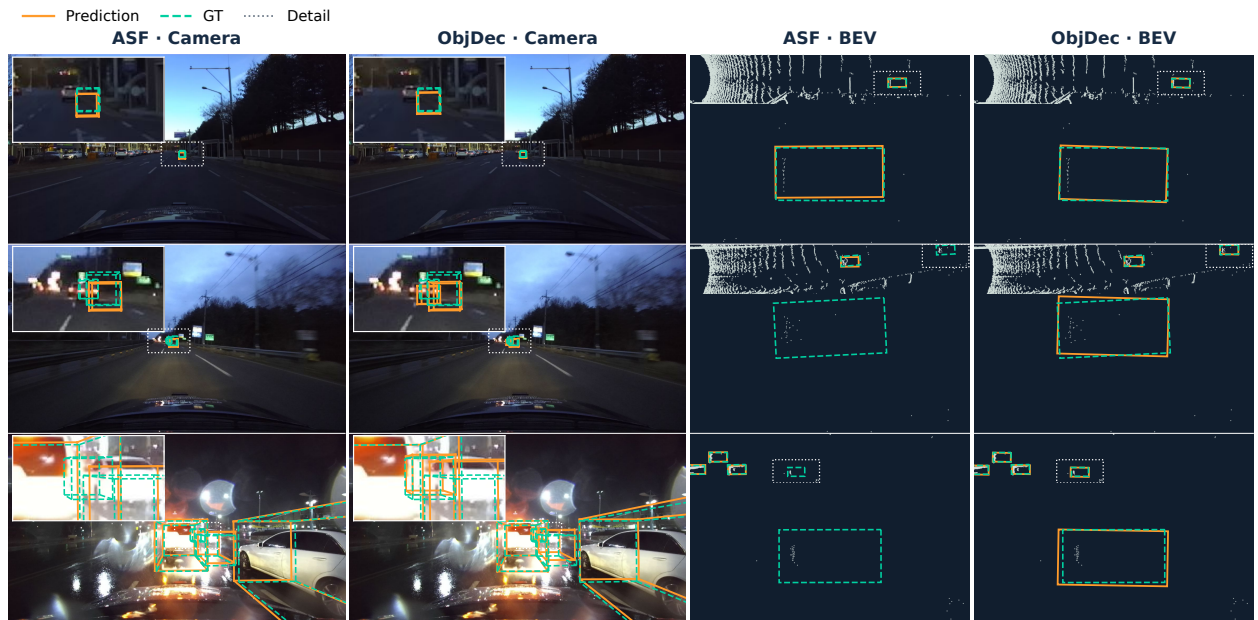
**Figure E.3:** Frame-level cross-modal correspondence controls using foreground-mean vectors in the original feature space. Same-frame pairs are compared with different-sequence pairs from the same weather group over ten fixed permutation seeds; matched scores use the same retained anchors. Weather-centered results subtract the mean for each modality, representation type, and weather group before computing cosine similarity. These are cosines of frame-mean vectors, unlike the averages of matched-patch cosines in Figure E.2. Seed variation concerns mismatch sampling, not independently trained models. High uncentered shared cosine persists under mismatching, so high similarity alone does not demonstrate semantic correspondence.



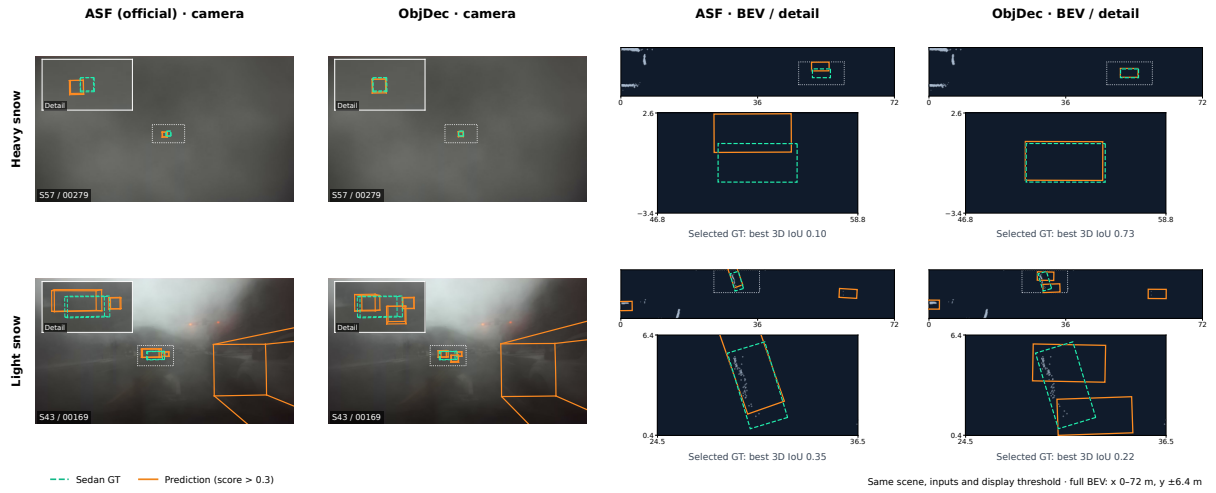
**Figure E.4:** Additional spatial-gate examples across seven weather groups. Each example shows the camera reference, LiDAR BEV, the predicted gate, and the identical gate with GT references added after prediction. All gates use the native  $16 \times 90$  grid, 0.8 m patches, and a common  $[0,1]$  color scale without smoothing or per-frame normalization. Gates are predicted from input features; GT supplies only training labels and post-hoc visualization or statistics. These selected examples do not estimate weather-wide performance. Table E.3 reports their frame-level statistics.



**Figure E.5:** Full-test foreground/background gate statistics over 10,065 frames. Left: weather-group means of per-frame foreground and background gate averages, with frames weighted equally. Right: distributions of each frame's foreground-minus-background mean, with outliers hidden in the display. GT-defined regions are applied after feature-based gate prediction. Across all frames, foreground/background means are 0.2701/0.1176, and 99.06% of frames have a larger foreground mean. These observational weather groups differ in sequences and scene composition and do not isolate a causal weather effect.



**Figure E.6:** Qualitative comparison of the released ASF checkpoint and ObjDec on identical K-Radar v1 test frames. The first two columns show calibrated camera projections with enlarged insets; the last two show full LiDAR BEV views above matched detail windows. Orange solid boxes denote Sedan predictions with scores above 0.3; green dashed boxes denote GT. Both methods use identical inputs, detection postprocessing, display thresholds, and camera/BEV crop windows. White dotted rectangles mark the enlarged regions. Full BEV views cover  $x \in [0, 72]$  m and  $y \in [-6.4, 6.4]$  m, while each detail window spans  $12 \times 6$  m. For the highlighted GT, maximum 3D IoU among retained predictions changes from ASF to ObjDec as follows, from top to bottom: 0.572 to 0.719, 0.000 to 0.476, and 0.000 to 0.741. These values describe selected-object box overlap rather than AP; the middle example remains below IoU 0.5. Figure E.7 includes an additional improvement and a localization counterexample.



**Figure E.7:** Additional paired detections from the released ASF checkpoint and ObjDec. Green dashed boxes are GT and orange solid boxes are predictions with  $\text{score} > 0.3$ ; both methods use identical camera projections and detail windows. The selected heavy-snow object's maximum geometric 3D IoU increases from 0.095 to 0.728, whereas the light-snow object's decreases from 0.354 to 0.223. The latter also appears in the seven-weather appendix gate visualization, where locally high foreground response coexists with poorer localization. Foreground-related activation does not replace final box regression. Displayed IoUs describe selected-object overlap rather than AP.

## F Sensor Availability and Corrupted Inputs

### F.1 Two input interventions

When a modality is removed, only the remaining sensor features are supplied to fusion, and gating, modality contributions, and context are recomputed for that set. Corrupted inputs instead retain the corresponding branch.  $C^*$  replaces the raw RGB image with a black frame before the original normalization and camera encoder.  $L^*$  uses a no-return convention: height-compressed sparse LiDAR features are zeroed before the original dense BEV backbone. This avoids sparse operators that do not support empty inputs; its precise interpretation is a no-return simulation at that feature interface.

All ten settings use one fixed checkpoint within each dataset version, without retraining or checkpoint selection per combination. Tables F.1 and F.2 report the full v1 ObjDec and v2 Strong, respectively. Unlike the SCL path in Section A.3, which reuses full-input controlled tokens, missing-sensor inference recomputes fusion control from the features actually available.

### F.2 Threshold effects, sensor dependence, and the ASF reference

Beyond the complete-input and two-sensor combinations discussed in the main text, Tables F.1 and F.2 add single-sensor inputs, corrupted inputs, and the second confidence setting. On v2,  $C+L^*+R$  obtains mean 3D@0.3 of 27.99 at confidence 0.3 and 34.70 at confidence 0.0. Removing LiDAR,  $C+R$ , yields 34.25 and 34.73 at those thresholds. Retaining a corrupted branch and removing that branch are not always equivalent across operating points.

Black-frame  $C^*+L+R$  remains close to complete-input performance, while camera-only predictions are weak. This combination establishes limited sensitivity to that particular camera-input change; a low camera contribution could also produce such behavior. The substantial decline without LiDAR indicates continued sensor dependence. If AP retention is reported, its denominator is complete-input AP for the same checkpoint and threshold; it is not interpreted as object recall.

ASF Table 3 (Paek and Kong, 2025) provides a reference for similar availability settings, but its precise corruption rules and camera-only evaluation scope are not fully matched to the local protocol. This appendix primarily compares input settings within fixed weights and does not use the starred rows for a strict cross-method robustness ranking.

**Table F.1: v1 input availability with fixed weights.** The same v1 ObjDec checkpoint is evaluated on 10,065 frames. The LR/conf=0.3 row uses the audited recomputation of saved predictions, reported to two decimals in its source. NR denotes an unverified summary for LR/conf=0.0, not zero performance. Other rows are read from their JSON records.

| Input  | conf | 3D@0.3 | 3D@0.5 | BEV@0.5 | $\Delta$ 3D@0.3 vs CLR |
|--------|------|--------|--------|---------|------------------------|
| C+L+R  | 0.3  | 88.36  | 67.50  | 88.10   | +0.00                  |
| C+L+R  | 0.0  | 88.06  | 72.83  | 87.18   | +0.00                  |
| L+R    | 0.3  | 88.06  | 71.68  | 85.45   | -0.30                  |
| L+R    | 0.0  | NR     | NR     | NR      | NR                     |
| C+L    | 0.3  | 86.77  | 72.58  | 86.35   | -1.59                  |
| C+L    | 0.0  | 86.16  | 69.74  | 83.76   | -1.90                  |
| C+R    | 0.3  | 57.62  | 34.59  | 55.52   | -30.74                 |
| C+R    | 0.0  | 63.65  | 38.48  | 60.28   | -24.41                 |
| L      | 0.3  | 79.58  | 60.05  | 77.96   | -8.77                  |
| L      | 0.0  | 85.46  | 59.23  | 77.96   | -2.60                  |
| R      | 0.3  | 49.81  | 26.64  | 46.80   | -38.55                 |
| R      | 0.0  | 61.75  | 30.86  | 52.60   | -26.31                 |
| C      | 0.3  | 0.86   | 0.16   | 0.21    | -87.49                 |
| C      | 0.0  | 0.32   | 0.03   | 0.14    | -87.74                 |
| C*     | 0.3  | 0.00   | 0.00   | 0.00    | -88.36                 |
| C*     | 0.0  | 0.17   | 0.01   | 0.07    | -87.89                 |
| C*+L+R | 0.3  | 88.36  | 67.41  | 88.09   | +0.00                  |
| C*+L+R | 0.0  | 88.06  | 72.75  | 87.17   | +0.00                  |
| C+L*+R | 0.3  | 50.26  | 34.83  | 49.27   | -38.09                 |
| C+L*+R | 0.0  | 63.05  | 38.48  | 61.12   | -25.01                 |

**Table F.2: v2 input availability with fixed Strong weights.** Equal class means over Sedan and Bus/Truck on 13,727 frames. Deltas use the full-input result at the same confidence filter. Stars denote the locally defined corruptions in Appendix F.

| Input  | conf | Mean 3D@0.3 | Mean 3D@0.5 | Mean<br>BEV@0.5 | $\Delta$ mean<br>3D@0.3 vs<br>CLR |
|--------|------|-------------|-------------|-----------------|-----------------------------------|
| C+L+R  | 0.3  | 66.91       | 43.05       | 60.93           | +0.00                             |
| C+L+R  | 0.0  | 69.41       | 43.81       | 63.23           | +0.00                             |
| L+R    | 0.3  | 66.02       | 41.81       | 59.93           | -0.89                             |
| L+R    | 0.0  | 68.96       | 42.77       | 62.74           | -0.46                             |
| C+L    | 0.3  | 64.08       | 41.17       | 57.80           | -2.83                             |
| C+L    | 0.0  | 65.13       | 41.16       | 58.53           | -4.28                             |
| C+R    | 0.3  | 34.25       | 16.99       | 29.97           | -32.66                            |
| C+R    | 0.0  | 34.73       | 16.82       | 29.89           | -34.68                            |
| L      | 0.3  | 59.39       | 32.36       | 53.45           | -7.53                             |
| L      | 0.0  | 62.33       | 33.47       | 54.75           | -7.08                             |
| R      | 0.3  | 30.51       | 13.59       | 26.28           | -36.40                            |
| R      | 0.0  | 31.63       | 13.45       | 26.70           | -37.79                            |
| C      | 0.3  | 0.01        | 0.00        | 0.01            | -66.90                            |
| C      | 0.0  | 0.01        | 0.00        | 0.01            | -69.40                            |
| C*     | 0.3  | 0.01        | 0.01        | 0.01            | -66.90                            |
| C*     | 0.0  | 0.01        | 0.01        | 0.00            | -69.40                            |
| C*+L+R | 0.3  | 66.93       | 42.98       | 60.91           | +0.01                             |
| C*+L+R | 0.0  | 69.40       | 43.73       | 63.19           | -0.01                             |
| C+L*+R | 0.3  | 27.99       | 14.11       | 25.01           | -38.92                            |
| C+L*+R | 0.0  | 34.70       | 15.51       | 30.76           | -34.71                            |

## G Cross-Dataset Architecture Adaptation

### G.1 Data processing and the common backbone

We use 8,391 training frames from the public package. The raw validation/test sets contain 1,498/1,501 frames and become 1,487/1,486 after deduplication. Car, Truck, and Bus are merged into Vehicle; the other classes are Pedestrian and Cyclist. ROI, difficulty, and IoU settings are listed in Table B.1. Local adaptation additionally includes ROI-based GT filtering, calibrated 2D box projection for difficulty filtering, and an intersection-area numerical correction. Dataset version, reporting split, and spatial coverage differ from the published benchmark tables (Yang et al., 2025); the results here extend the local architectural comparison defined in the main text.

The camera branch combines ResNet-50 multiscale features with a learned depth distribution and LSS (Phillion and Fidler, 2020) projection to BEV, using  $288 \times 512$  input images. LiDAR and radar use independent PillarVFE, scatter, and BEV convolutional stems, with four and six point fields, respectively. Each pillar contains at most 32 points, and at most 32,000 pillars are retained. Each modality outputs 64 channels. Fusion produces 128 channels before the common multiscale BEV backbone; the detection head receives 384 channels and uses anchors determined from training-set statistics.

Common encoder and detection-head tensors are copied from the same untrained initialization, without warm-starting from a trained target-dataset checkpoint. Augmentations include global rotation within  $\pm 0.392699$  radians, scaling within  $[0.95, 1.05]$ , and flipping with probability 0.5. Camera geometry and point clouds share the same spatial transform. Other training settings are provided in Section A.3.

### G.2 Fusion structures and spatial resolution

Concat applies a  $3 \times 3$  convolution, BatchNorm, and ReLU after channel-wise BEV concatenation, learning combinations across modalities and neighboring locations. ASF-style uses common projection, local attention, and PFT. ObjDec jointly incorporates the representation, gating, modality-contribution, and context paths from Section 3 into fusion. The three-sensor group shares encoders and a detection head. ObjDec-LR is trained with only LiDAR and radar branches from the outset; comparisons with L4DR retain their native architectural differences.

The 0.4 m and 0.16 m settings use BEV grids of  $256 \times 256$  and  $640 \times 640$ , respectively, with detection-head stride two relative to BEV. The earlier  $4 \times 4$  patch setting uses 16 queries, and the  $2 \times 2$  setting uses four; both output 128 channels. Physical patch coverage changes from 1.6 m or 0.8 m at a 0.4 m grid to 0.32 m at a 0.16 m grid. Grid and patch changes therefore alter both computation and the size of local fusion regions relative to objects.

### G.3 Supplementary results under the full training budget

The main Table 4 presents published references and local fusion comparisons; this section additionally reports the Concat control on the common backbone. Table G.1 retains the earlier 0.4 m group, selected epochs, and corresponding validation/test scores omitted from the main table. Table G.2 adds per-class BEV and last-epoch results for completed 0.16 m runs. At 0.4 m, changing ObjDec from  $4 \times 4$  to  $2 \times 2$  patches increases test Pedestrian 3D AP from 58.75 to 62.36 and mean 3D AP from 72.22 to 74.03. This change also modifies the query count. Concat obtains mean test 3D AP of 75.77 at 0.4 m and 81.90 at 0.16 m, showing that the common detection backbone also benefits from finer grids. Gains from finer resolution cannot all be attributed to ObjDec. At 0.16 m, validation-selected

ObjDec trails Concat by 0.52 and 0.20 percentage points in mean test 3D and BEV AP, respectively; these results do not establish superiority over all evaluated fusion architectures.

All five 0.16 m runs have completed 80 training epochs and final best/last evaluations. Validation-selected three-sensor ObjDec at epoch 70 obtains mean test 3D/BEV AP of 81.38/86.27, compared with 81.55/86.35 at epoch 80. ObjDec-LR obtains 79.65/83.81 at its selected epoch 75 and 79.75/83.80 at epoch 80. Last-checkpoint test differences do not change the validation-based selection rule. ASF-style and L4DR both select epoch 80. Relative to L4DR, ObjDec-LR improves class-wise 3D AP by 0.71/1.58/1.93 points and BEV AP by 0.29/1.73/1.73 points. Relative to ASF-style, three-sensor ObjDec improves Cyclist 3D and BEV AP by 0.95/1.21 points, with small decreases in Vehicle 3D and BEV AP. **[TO COMPLETE: complete 80-epoch validation curves for Figure G.1; all required validation checkpoints are available.]** Curves use actual validation checkpoints without combining per-class maxima from different models.

#### G.4 Additional architecture adaptation on VoD

**Setup.** We supplement the main V2X-Radar-V evaluation by training a LiDAR-4D-radar version within a native PointPillars (Lang et al., 2019) detection backbone on VoD (Palffy et al., 2022), using 5,139 training and 1,296 validation frames. Inputs comprise one LiDAR frame and five radar frames. Both recent ObjDec runs use a 0.16 m grid, a  $320 \times 320$  BEV,  $2 \times 2$  patches, and four queries. Each modality has two 64-channel  $3 \times 3$  Conv-BN-ReLU layers for local encoding, followed by token LayerNorm. Both models are trained from scratch for 80 epochs with seed 666, FP32, Adam OneCycle, and a peak learning rate of 0.003. A microbatch of eight with two accumulation steps gives an effective batch size of 16. Full validation is performed every five epochs; checkpoints are selected by official EAA mean AP and used unchanged for DC and other metrics. The score and NMS thresholds remain 0.1 and 0.01.

**Anchor adaptation.** The second run changes only class-specific anchor dimensions and bottom heights. Priors are computed from training annotations before augmentation, transformed into LiDAR coordinates and filtered using the native training ROI’s box-center rule. Dimension and bottom-height medians are rounded to 0.01 m, without fitting to validation annotations. Length/width/height values for Car, Pedestrian, and Cyclist are 4.19/1.81/1.52, 0.65/0.63/1.69, and 1.94/0.78/1.76 m, with bottom heights of  $-1.65$ ,  $-1.62$ , and  $-1.68$  m. This comparison evaluates the joint change in dimensions and heights.

**Results.** Table G.3 (a,b) use official VoD region/class filtering and 11-point 3D AP, with IoU thresholds of 0.5/0.25/0.25 for Car/Pedestrian/Cyclist. EAA and DC denote the Entire Annotated Area and Driving Corridor; both report 3D detection. The  $2 \times 2$ -patch model with local encoding selects epoch 75 and obtains EAA/DC of 71.23/84.69. Adapted anchors select epoch 80 and yield 72.03/83.76. The 0.80-point EAA gain is primarily associated with a 4.51-point Cyclist improvement, while DC decreases by 0.93 points. Relative to the local L4DR reference, EAA is 1.03 points higher and DC is 1.08 points lower; aggregate EAA/DC remain below the published L4DR results (Huang et al., 2025). Table G.4 additionally reports KITTI AP\_R40 for the same checkpoints: Moderate mean 3D AP changes from 78.64 to 78.35, whereas BEV AP increases from 80.55 to 80.92, showing a tradeoff across metrics.

**Comparison scope.** Historical PP-Concat uses AMP and a physical batch size of 16, and the earlier ObjDec is initialized from Concat. Local L4DR uses 100 epochs, two GPUs, and SyncBN; its epoch-99 checkpoint is the better of the evaluated late-stage candidates. These rows provide historical and external architectural references with differing training conditions. A Concat reference with the same local encoding and training settings is not yet available, so the combined gains over the historical baseline cannot be attributed entirely to representation decoupling. These results

**Table G.1: Spatial granularity and full-budget results.** Best checkpoints are selected by validation under an 80-epoch budget, using Moderate AP\_R40. The 0.4 m group supplements the main 0.16 m comparison. Validation and test results are reported separately.

| Method        | Sensors | Grid (m) | Best epoch | Val mean 3D | Test Vehicle | Test Pedestrian | Test Cyclist | Test mean 3D | Test mean BEV |
|---------------|---------|----------|------------|-------------|--------------|-----------------|--------------|--------------|---------------|
| Concat        | C+L+R   | 0.4      | 75         | 74.31       | 81.52        | 65.81           | 79.98        | 75.77        | 81.26         |
| ASF-style     | C+L+R   | 0.4      | 75         | 72.76       | 78.57        | 58.93           | 79.02        | 72.17        | 78.21         |
| ObjDec 4×4    | C+L+R   | 0.4      | 80         | 71.31       | 78.82        | 58.75           | 79.09        | 72.22        | 78.76         |
| ObjDec 2×2    | C+L+R   | 0.4      | 80         | 74.19       | 80.73        | 62.36           | 78.99        | 74.03        | 80.26         |
| Concat        | C+L+R   | 0.16     | 75         | 81.46       | 84.81        | 76.84           | 84.05        | 81.90        | 86.47         |
| ASF-style     | C+L+R   | 0.16     | 80         | 81.03       | 84.54        | 76.17           | 83.18        | 81.30        | 85.72         |
| ObjDec 2×2    | C+L+R   | 0.16     | 70         | 81.41       | 84.06        | 75.94           | 84.13        | 81.38        | 86.27         |
| L4DR          | L+R     | 0.16     | 80         | 77.19       | 83.17        | 73.25           | 78.31        | 78.24        | 82.56         |
| ObjDec-LR 2×2 | L+R     | 0.16     | 75         | 79.12       | 83.87        | 74.84           | 80.24        | 79.65        | 83.81         |

support architecture adaptation after training on VoD, without establishing zero-shot transfer or overall state-of-the-art performance on VoD.

**Table G.2: BEV and last-checkpoint results for completed 0.16 m runs.** Deduplicated test results. Best is selected by validation, whereas last is epoch 80. Reporting last does not change the selection rule. Best/last evaluations are complete for all five 0.16 m runs.

| Method        | Checkpoint | Epoch | BEV<br>Vehicle | BEV<br>Pedes-<br>trian | BEV<br>Cyclist | Mean<br>BEV | Mean 3D |
|---------------|------------|-------|----------------|------------------------|----------------|-------------|---------|
| Concat        | best       | 75    | 92.12          | 80.66                  | 86.64          | 86.47       | 81.90   |
| Concat        | last       | 80    | 92.02          | 80.70                  | 86.69          | 86.47       | 81.97   |
| ASF-style     | best       | 80    | 90.95          | 80.41                  | 85.80          | 85.72       | 81.30   |
| ASF-style     | last       | 80    | 90.95          | 80.41                  | 85.80          | 85.72       | 81.30   |
| ObjDec 2×2    | best       | 70    | 90.91          | 80.88                  | 87.02          | 86.27       | 81.38   |
| ObjDec 2×2    | last       | 80    | 90.84          | 81.16                  | 87.07          | 86.35       | 81.55   |
| L4DR          | best       | 80    | 89.74          | 76.60                  | 81.35          | 82.56       | 78.24   |
| L4DR          | last       | 80    | 89.74          | 76.60                  | 81.35          | 82.56       | 78.24   |
| ObjDec-LR 2×2 | best       | 75    | 90.02          | 78.34                  | 83.08          | 83.81       | 79.65   |
| ObjDec-LR 2×2 | last       | 80    | 90.10          | 78.21                  | 83.09          | 83.80       | 79.75   |

**Table G.3: VoD region-based 3D detection results.** All methods use L+R without added synthetic fog. Official VoD 11-point 3D AP uses class IoUs of 0.5/0.25/0.25. Local results use 1,296 validation frames, with the same checkpoint for each method across EAA and DC. The two recent ObjDec runs select checkpoints by EAA; training and historical-reference differences are detailed in Section G.4. Reported rows are from Table 3 of the published L4DR paper (Huang et al., 2025), with means computed from its class values. Local means retain the evaluator output.

| (a) Entire Annotated Area (EAA) |          |       |       |                 |         |       |
|---------------------------------|----------|-------|-------|-----------------|---------|-------|
| Method                          | Source   | Epoch | Car   | Pedes-<br>trian | Cyclist | Mean  |
| PP-Concat (historical)          | Local    | 80    | 66.88 | 63.38           | 79.39   | 69.88 |
| ObjDec (warm-start, historical) | Local    | 79    | 67.50 | 63.66           | 79.38   | 70.18 |
| ObjDec ( $2 \times 2$ + local)  | Local    | 75    | 68.72 | 65.59           | 79.38   | 71.23 |
| ObjDec (+ adapted anchors)      | Local    | 80    | 66.93 | 65.28           | 83.89   | 72.03 |
| L4DR                            | Local    | 99    | 68.69 | 65.35           | 78.98   | 71.00 |
| InterFusion                     | Reported | —     | 66.50 | 64.50           | 78.50   | 69.83 |
| L4DR                            | Reported | —     | 69.10 | 66.20           | 82.80   | 72.70 |

| (b) Driving Corridor (DC)       |          |       |       |                 |         |       |
|---------------------------------|----------|-------|-------|-----------------|---------|-------|
| Method                          | Source   | Epoch | Car   | Pedes-<br>trian | Cyclist | Mean  |
| PP-Concat (historical)          | Local    | 80    | 90.77 | 71.09           | 89.53   | 83.80 |
| ObjDec (warm-start, historical) | Local    | 79    | 90.59 | 71.02           | 89.76   | 83.79 |
| ObjDec ( $2 \times 2$ + local)  | Local    | 75    | 90.74 | 73.23           | 90.09   | 84.69 |
| ObjDec (+ adapted anchors)      | Local    | 80    | 89.44 | 73.02           | 88.83   | 83.76 |
| L4DR                            | Local    | 99    | 90.51 | 75.13           | 88.90   | 84.84 |
| InterFusion                     | Reported | —     | 90.70 | 72.00           | 88.70   | 83.80 |
| L4DR                            | Reported | —     | 90.80 | 76.10           | 95.50   | 87.47 |

**Table G.4: Supplementary KITTI difficulty metrics on VoD.** Equal class means for the same EAA-selected checkpoints as Table G.3, using 3D/BEV IoUs of 0.5/0.25/0.25. KITTI difficulty filtering and AP\_R40 differ from the region filtering and 11-point AP of official EAA/DC, so the two sets of scores are not interchangeable.

| Configuration                  | Epoch | Metric | Easy mean | Moderate<br>mean | Hard mean |
|--------------------------------|-------|--------|-----------|------------------|-----------|
| ObjDec ( $2 \times 2$ + local) | 75    | 3D     | 83.43     | 78.64            | 72.23     |
| ObjDec (+ adapted anchors)     | 80    | 3D     | 83.06     | 78.35            | 72.02     |
| ObjDec ( $2 \times 2$ + local) | 75    | BEV    | 85.47     | 80.55            | 74.56     |
| ObjDec (+ adapted anchors)     | 80    | BEV    | 85.33     | 80.92            | 75.02     |

## H Efficiency Measurement and Scope

### H.1 Paired timing protocol

Timing uses an RTX A6000, batch size 1, four PyTorch CPU threads, FP32 tensors, and no autocast. The original environment’s matmul/cuDNN TF32 settings remain enabled; cuDNN deterministic is enabled and benchmark is disabled. The PyTorch version is 1.10.1+cu113. We sample 120 test indices uniformly, using the first 20 for warmup and the remaining 100 for timing, and repeat the same frames in two rounds.

Each frame’s CPU input is copied for the eager and Graph paths, alternating their order and synchronizing each measurement. Model order is ObjDec, ASF, ASF, ObjDec. Timing covers the network call, including point-cloud preprocessing, host-to-device transfer, encoding, fusion, decoding, NMS, and GT recall computation in the original evaluation path. Disk access, collation, initial graph capture, initialization, and result verification are excluded. Training on other GPUs remains active, so host load is not fully isolated. Table H.1 adds median, P95, and timing standard deviation; very small mean-latency differences are not interpreted as stable architectural advantages.

### H.2 Graph replay and numerical checks

CUDA Graph captures only the repeated camera-Swin and fusion computations. Each frame supplies new images and BEV features and executes the captured operators; calibration is read per frame, and predictions are not cached across frames. Inputs with different shapes, sensor sets, or visualization requirements fall back to eager execution according to the implementation conditions.

The paired runs checked 240 forwards for each model. Under identical captured inputs, all capture checks were exact. Across the full paths, all checked tensors were exactly equal in 218/240 cases for each model; repeated eager checks also showed variation in the unmodified radar path. Label arrays matched in 238/240 cases for ObjDec and 235/240 for ASF. These sampled checks therefore do not establish complete numerical equivalence of the full inference pipeline.

All four timing runs encountered a native exit error after their result files had been written. An independent read verified the stored timing and comparison records; the original profiling path exhibited a similar exit issue. The two visualization inference jobs likewise completed their exports before native exit errors, and their output files were checked independently. Full-test AP has not been reevaluated for the Graph path. We therefore report measured latency and sampled checks, without claiming fully validated deployment or unchanged full-set detection accuracy.

### H.3 Scope of the evidence

The experiments use the current single-seed training and checkpoint-selection procedures, with part of the v1 preliminary-selection record still to be documented. Strong on v2 does not contain the complete object-context path. V2X and VoD experiments train on their target datasets and establish adaptation under the stated protocols, rather than zero-shot transfer. Sensor-availability results retain weak camera-only performance and a marked dependence on LiDAR. Finally, cosine constraints and foreground supervision organize representations but do not establish an identifiable semantic decomposition, statistical independence, or an exact reconstruction identity  $c + u = t$ . These boundaries delimit what the present ablations and visualizations support.

**Table H.1: Latency distributions.** Each model/path uses 100 distinct timed frames repeated twice. The final column is the reciprocal of measured latency, excluding disk loading. Standard deviations describe timing variation rather than multi-seed detection uncertainty.

| Model  | Execution  | Mean ms | Median ms | P95 ms | Std ms | 1000 /<br>mean ms |
|--------|------------|---------|-----------|--------|--------|-------------------|
| ASF    | Eager      | 84.00   | 82.63     | 93.26  | 5.34   | 11.91             |
| ASF    | CUDA Graph | 59.30   | 59.10     | 64.82  | 2.83   | 16.86             |
| ObjDec | Eager      | 84.69   | 84.50     | 89.98  | 3.01   | 11.81             |
| ObjDec | CUDA Graph | 59.57   | 59.23     | 65.93  | 2.92   | 16.79             |